



哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY

立足航天，服务国防，面向国民经济主战场



《计算机网络》

第7章 无线与移动网络

聂兰顺



第7章 无线与移动网络

本章学习目标

- ❖ 理解无线网络组成要素
- ❖ 理解无线链路与无线网络特性
- ❖ 掌握IEEE802.11无线局域网体系结构
- ❖ 掌握CSMA/CA协议
- ❖ 了解移动通信网络体系结构
- ❖ 掌握实现网络移动性的基本原理
- ❖ 理解移动IP原理与通信过程
- ❖ 了解移动通信网络移动性原理

主要内容

- ❖ 7.1 无线与移动网络概述
- ❖ 7.2 无线链路与网络特性
- ❖ 7.3 无线局域网：IEEE802.11
- ❖ 7.4 移动通信网络体系结构
- ❖ 7.5 网络移动管理基本原理
- ❖ 7.6 移动IP
- ❖ 7.7 移动通信网络的移动管理
- ❖ 7.8 无线与移动对高层协议的影响





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7.1 无线与移动网络概述

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无线和移动网络：背景

7.1 无线与移动网络概述

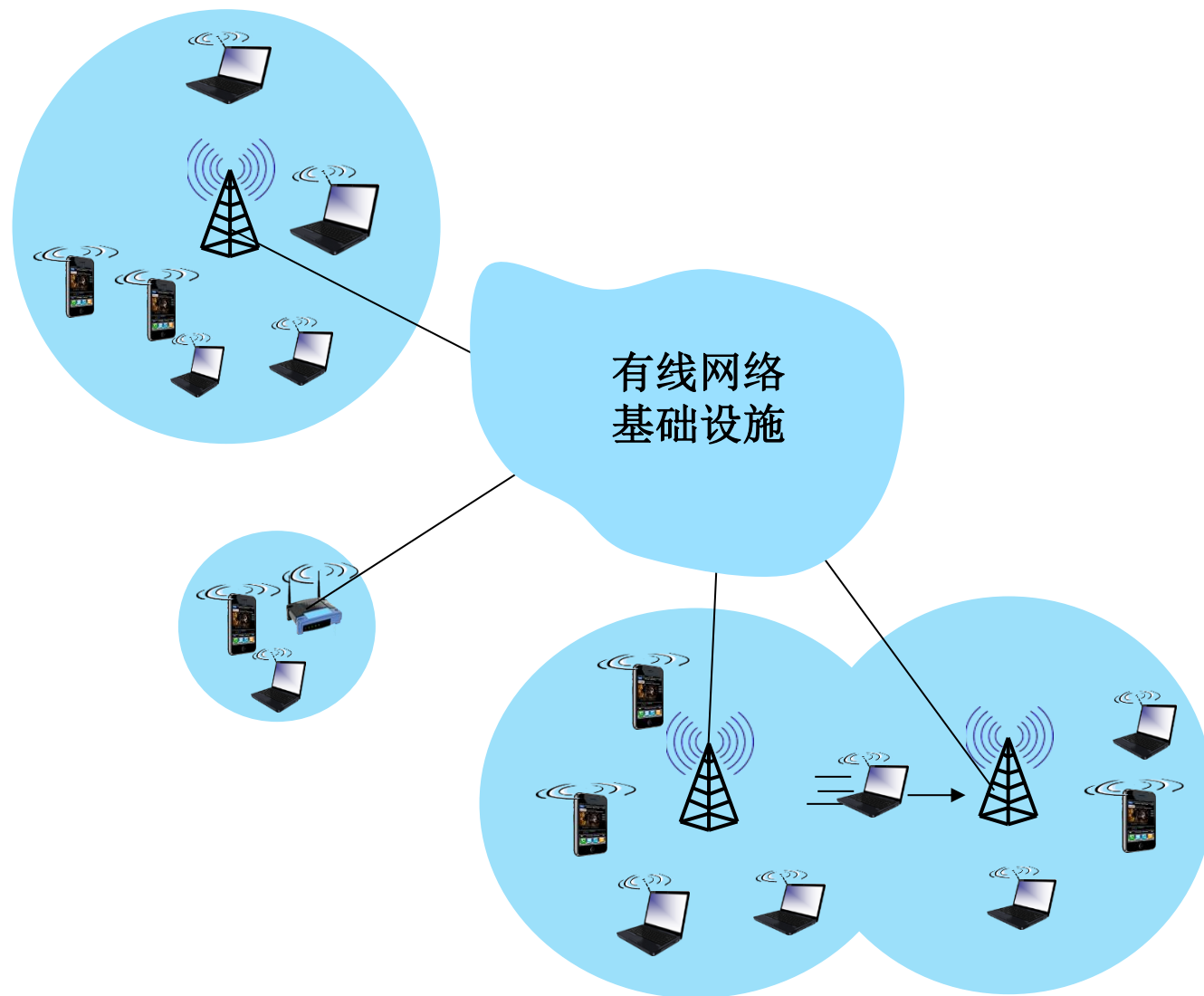
- ❖ 无线（移动）电话用户多于固定（有线）电话用户（2019年为10:1）！
- ❖ 移动宽带连接设备多于固定宽带连接设备（2019年为5:1）！
 - 4G/5G蜂窝网络现在采用互联网协议栈，包括SDN技术
- ❖ 两个重要（但不同）的挑战
 - 无线:通过无线链路进行通信
 - 移动:处理改变网络接入点的移动用户





无线网络的组成要素

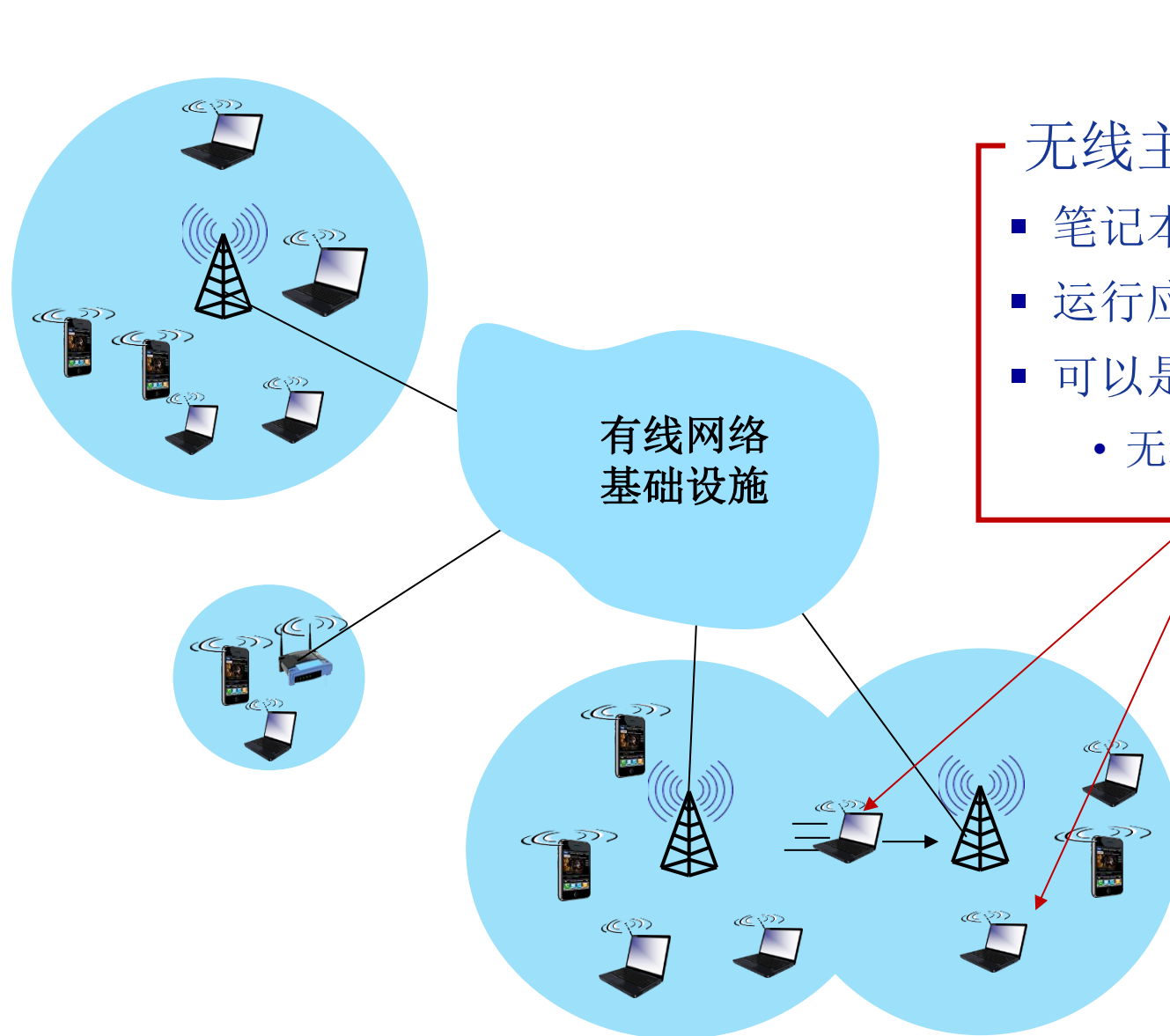
7.1 无线与移动网络概述





无线网络的组成要素

7.1 无线与移动网络概述



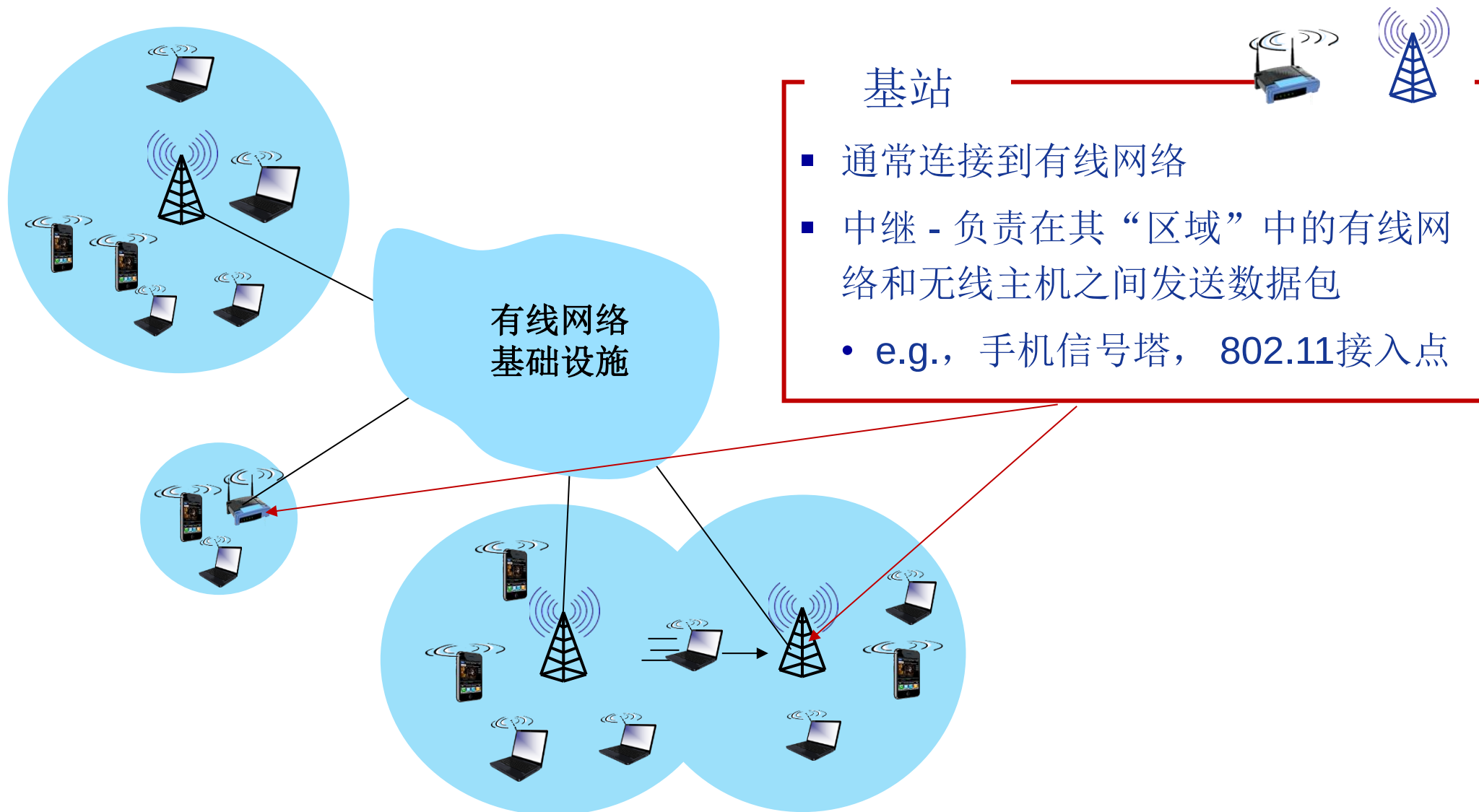
无线主机

- 笔记本电脑、智能手机、IoT
- 运行应用程序
- 可以是静止的（非移动）或移动的
 - 无线并不总是意味着移动性!



无线网络的组成要素

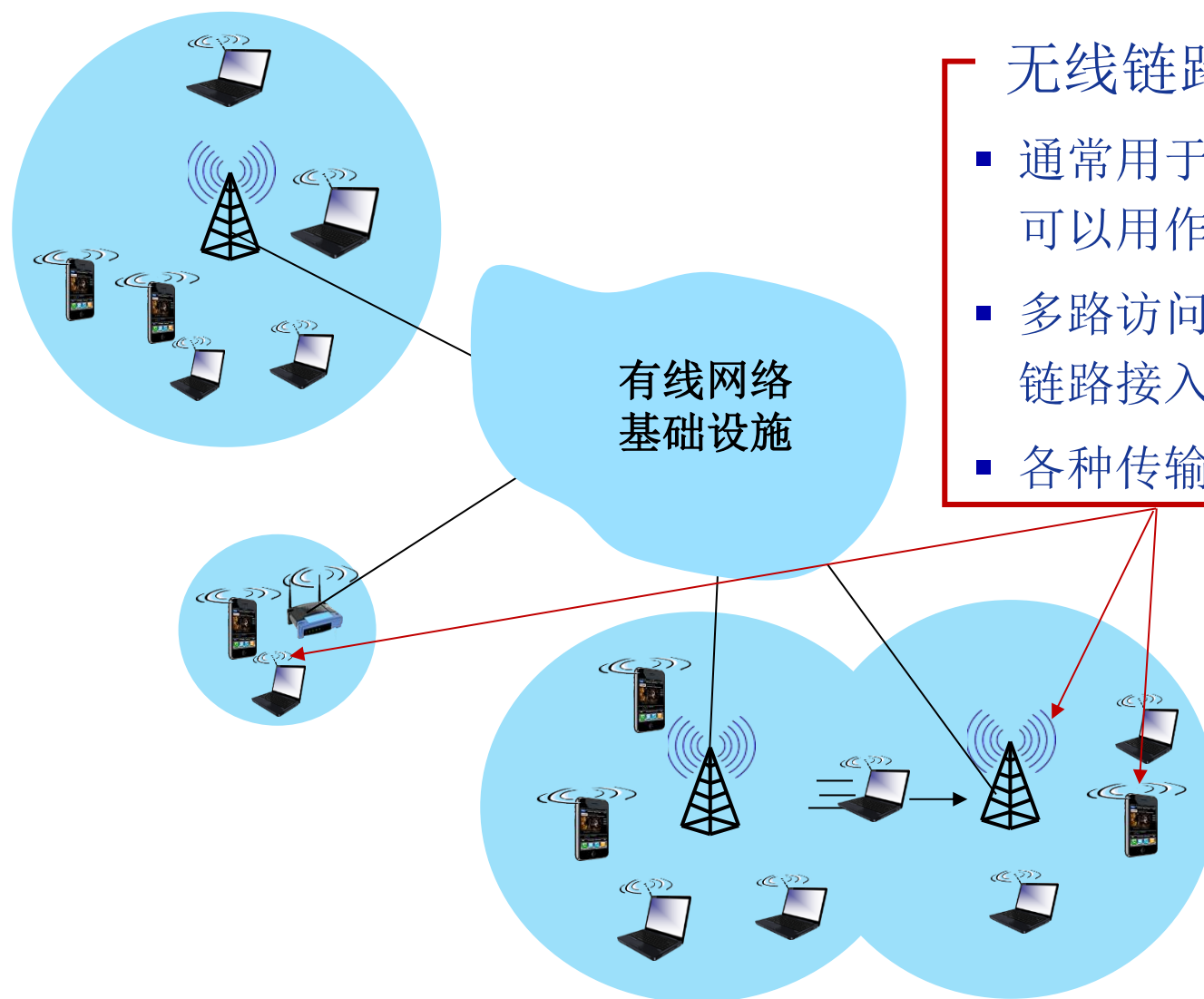
7.1 无线与移动网络概述





无线网络的组成要素

7.1 无线与移动网络概述



无线链路



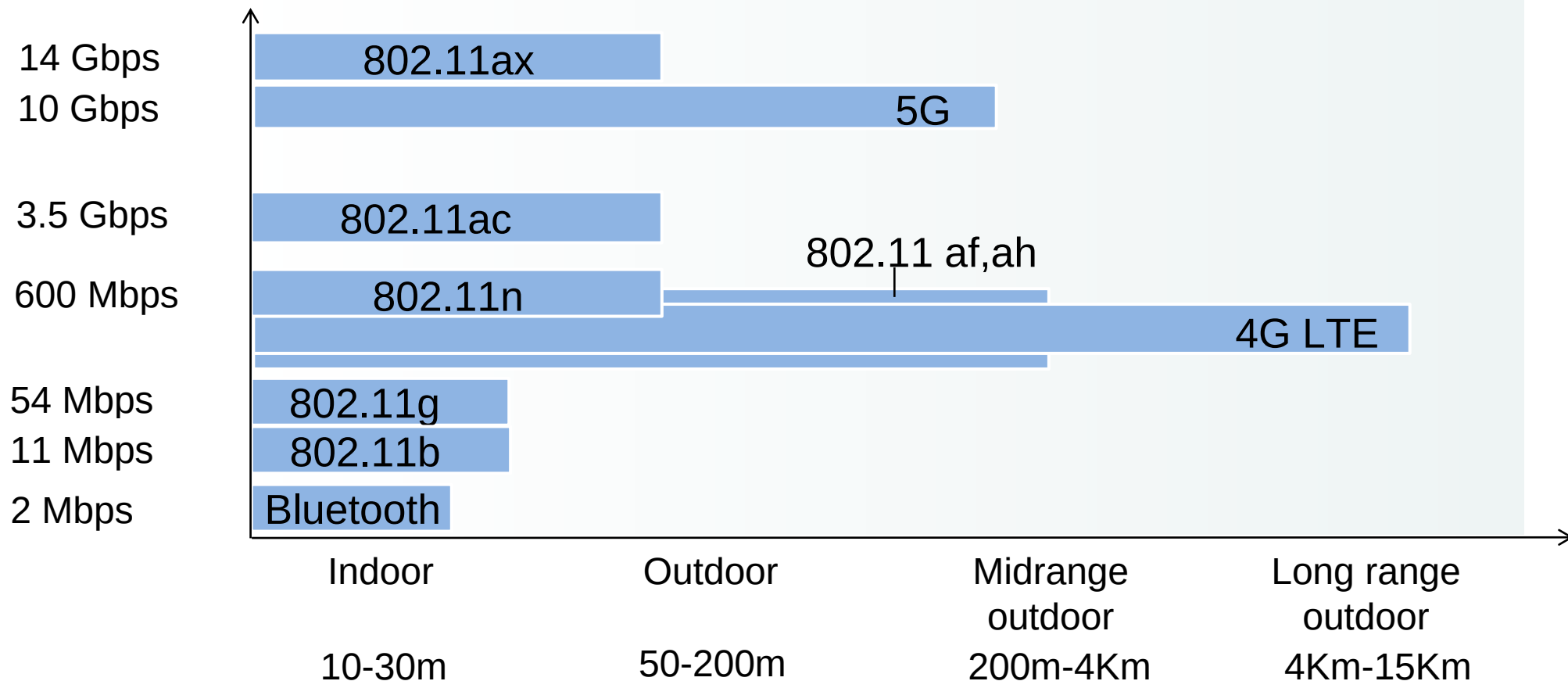
- 通常用于将移动设备连接到基站，也可以用作骨干链路
- 多路访问控制协议（MAC）协议协调链路接入
- 各种传输速率、距离和频段





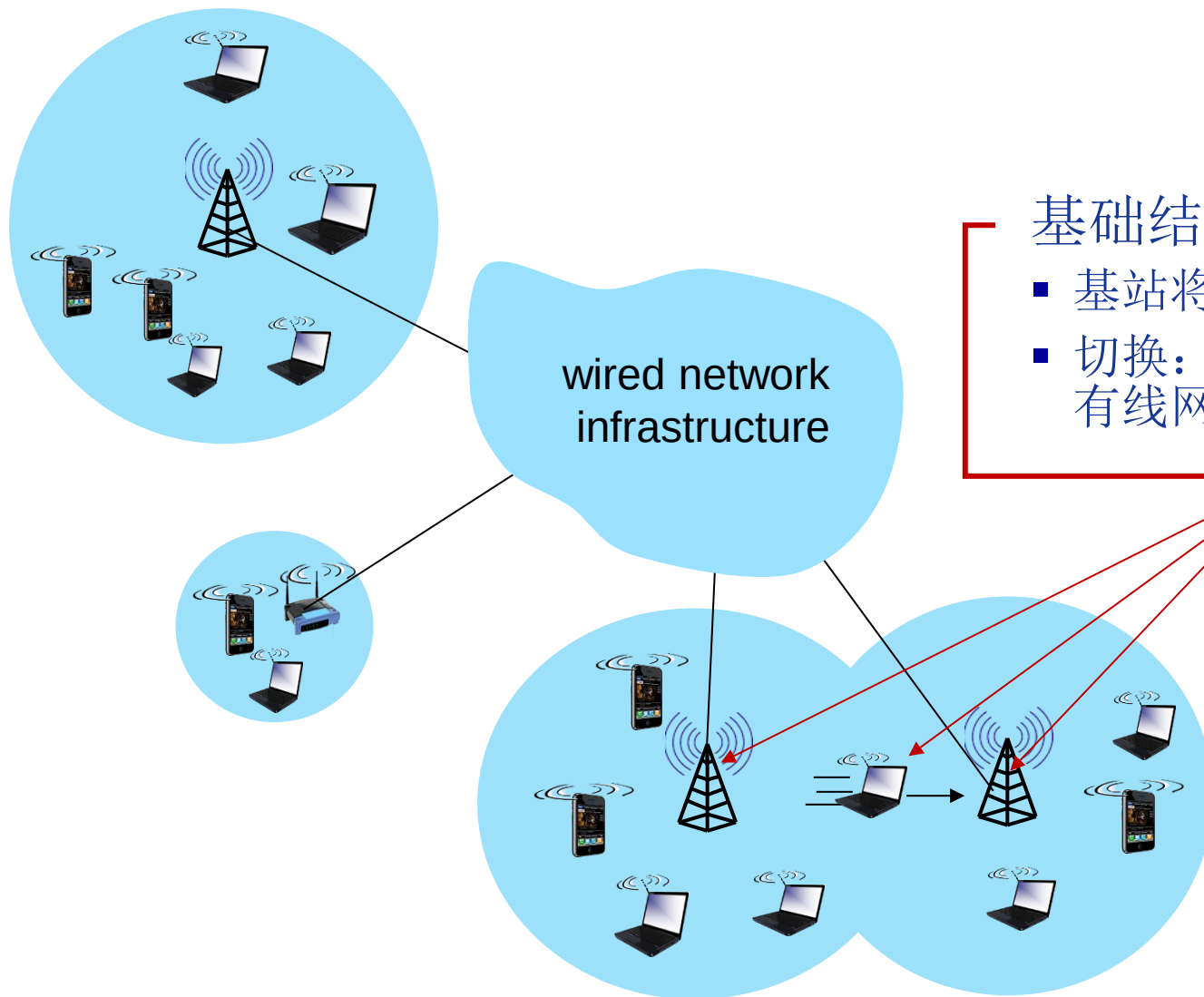
典型无线网络

7.1 无线与移动网络概述



无线网络的模式

7.1 无线与移动网络概述



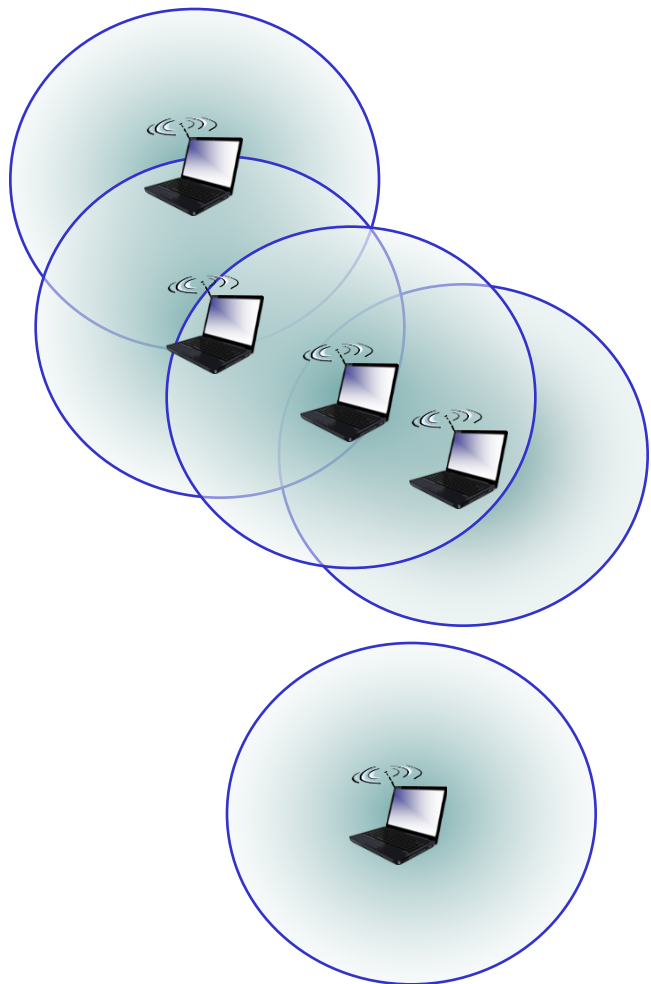
基础结构模式

- 基站将移动终端连接到有线网络
- 切换：移动终端更换基站，提供有线网络连接



无线网络的模式

7.1 无线与移动网络概述



特定网(ad hoc)模式

- 无基站
- 结点只能传输到无线链路覆盖范围内的其他结点
- 结点将自己组织成一个网络（自组网）：在结点之间路由





Wireless network taxonomy

7.1 无线与移动网络概述

	单跳	多跳
基础设施 (e.g., APs)	主机连接到基站（WiFi，蜂窝），基站连接到更大的互联网	主机可能必须通过多个无线节点中继才能连接到更大的互联网：网状网络
无基础设施	没有基站，无法连接到更大的互联网（蓝牙，点对点网络）	没有基站，没有连接到更大的互联网。可能要中继才能到达其他给定的无线节点MANET，VANET





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7.2 无线链路与网络特性

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无线链路特性(1)

7.1 无线与移动网络概述

7.2 无线链路与网络特性

与有线链路的重要区别....

- **信号强度衰减**：无线电信号在介质中传播时衰减（路径损耗）
- **来自其他来源的干扰**：许多设备（例如 WiFi、蜂窝网络、电机）共享无线网络频率（例如 **2.4 GHz**）：干扰
- **多路径传播**：无线电信号反射自物体表面，在不同时刻到达目的地



...使通过（甚至是点对点）无线链路进行通信变得更加“困难”



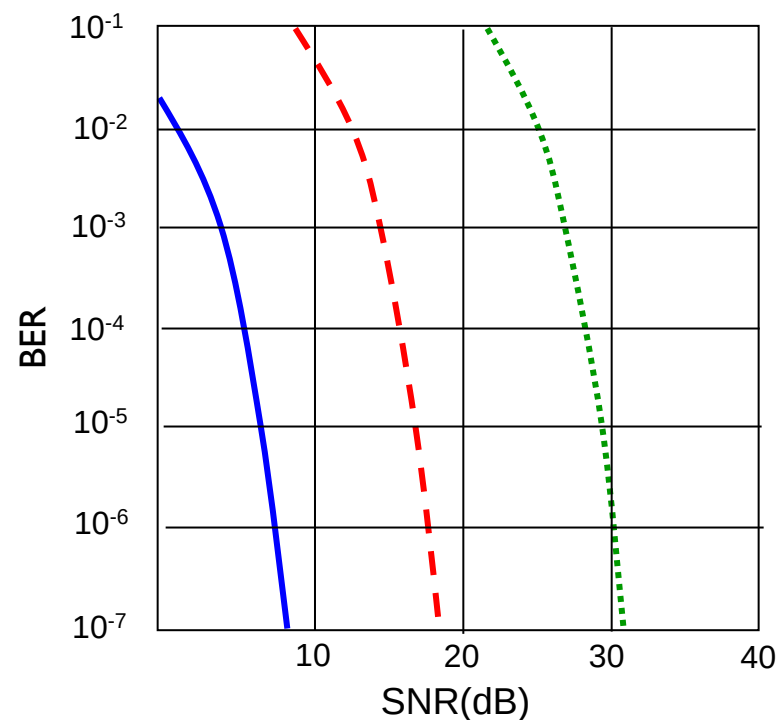


无线链路特性(2)

7.1 无线与移动网络概述

7.2 无线链路与网络特性

- SNR: 信噪比
 - 较大SNR – 更容易从噪声中提取信号
- SNR vs BER间的权衡
 - 给定物理层: 增加功率 → 增加SNR → 降低 BER
 - 给定SNR: 选择满足 BER 要求的物理层, 提供最高的吞吐量
 - SNR 可能会随着移动性而变化: 动态适应物理层 (调制技术、速率)



- QAM256 (8 Mbps)
- - - QAM16 (4 Mbps)
- BPSK (1 Mbps)



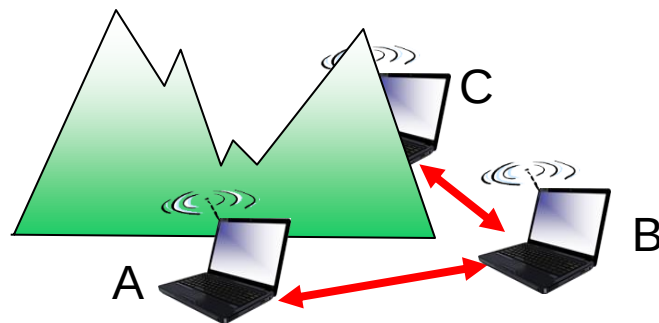


无线链路特性(3)

7.1 无线与移动网络概述

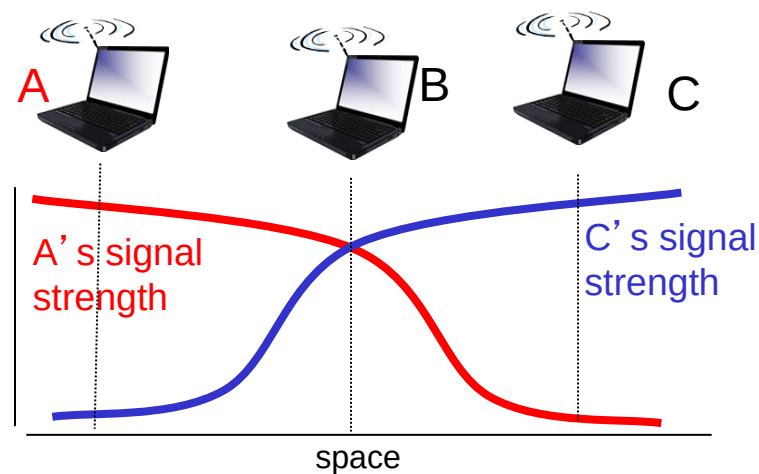
7.2 无线链路与网络特性

多个无线发送器、接收器会产生其他问题（超出多路访问控制）：



隐藏终端问题：

- B、A彼此可以侦听到
- B、C彼此可以侦听到
- A、C彼此侦听不到，即A和C不知道他们的信号在B处会相互干扰



信号衰减：

- B、A彼此可以侦听到
- B、C彼此可以侦听到
- A、C 由于信号衰减，彼此侦听不到，在B处会相互干扰





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7.3 无线局域网：IEEE802.11

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IEEE 802.11无线局域网

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

IEEE 802.11 standard	Year	Max data rate	Range	Frequency
802.11b	1999	11 Mbps	30 m	2.4 Ghz
802.11g	2003	54 Mbps	30m	2.4 Ghz
802.11n (WiFi 4)	2009	600	70m	2.4, 5 Ghz
802.11ac (WiFi 5)	2013	3.47Gpbs	70m	5 Ghz
802.11ax (WiFi 6)	2020 (exp.)	14 Gbps	70m	2.4, 5 Ghz
802.11af	2014	35 – 560 Mbps	1 Km	unused TV bands (54-790 MHz)
802.11ah	2017	347Mbps	1 Km	900 Mhz

- 所有都使用 CSMA/CA 进行多址访问，并具有基站和点对点网络版本



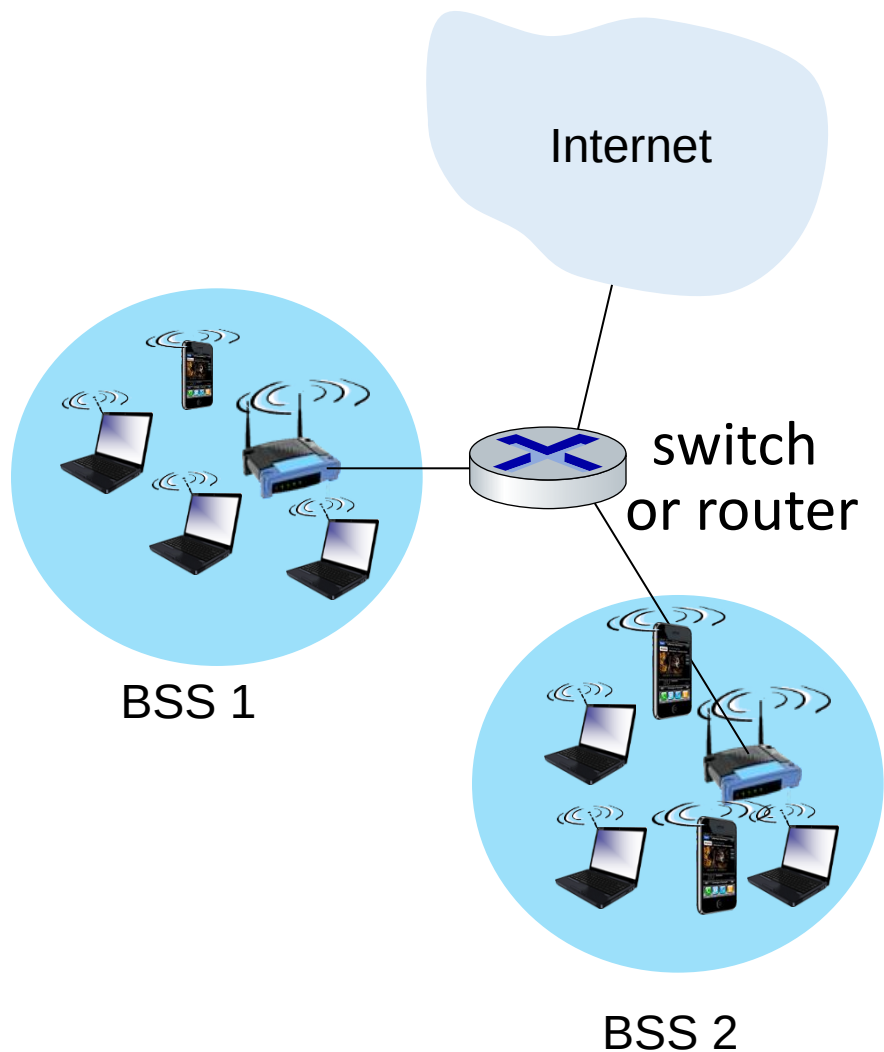


802.11 LAN体系结构

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11



❖ 无线主机与基站通信

- 基站(base station) = 访问点(access point-AP)

❖ 基本服务集BSS(Basic Service Set), 也称为单元(cell)

- 基础设施网络模式:
 - 无线主机
 - AP: 基站
- 自组网(ad hoc)模式:
 - 只有主机





802.11:信道与关联

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

- 频谱划分为不同频率的信道
 - 每个AP选择一个频率(信道)
 - 存在干扰可能: 相邻的AP可能选择相同的信道!
- 主机: 必须与某个AP**关联(associate)**
 - 扫描信道, 监听包含AP名称(服务集标识符-SSID)和MAC地址的**信标(beacon)**帧
 - 选择一个AP进行关联
 - 可能需要进行身份认证
 - 典型情形: 运行DHCP获取IP地址等信息



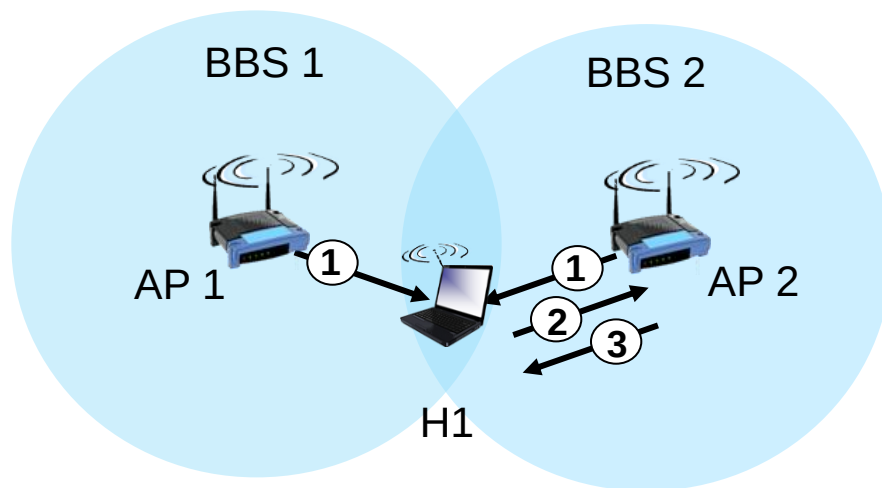


802.11关联:被动扫描与主动扫描

7.1 无线与移动网络概述

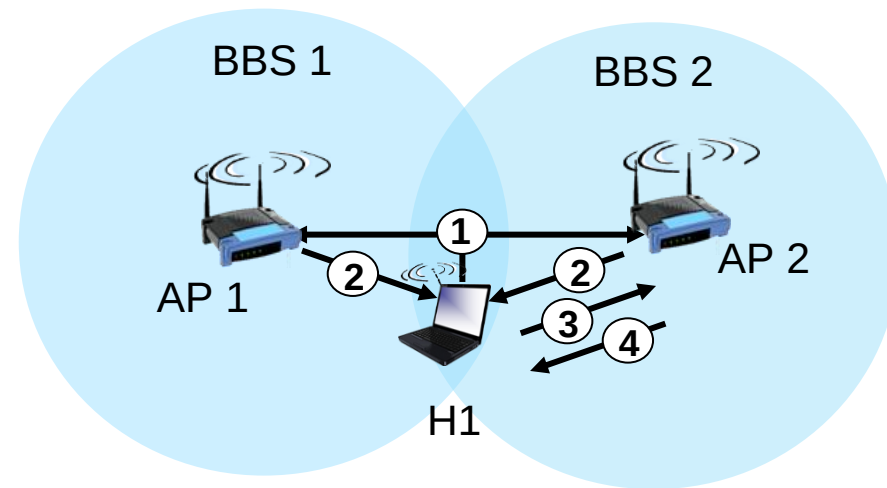
7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11



被动扫描(scanning):

- (1) 各AP发送信标帧
- (2) 主机(H1)向选择的AP发送关联请求帧
- (3) AP向主机(H1)发送关联响应帧



主动扫描:

- (1) 主机(H1)主动广播探测请求帧 (Probe Request Frame)
- (2) AP发送探测响应帧 (Probe Response Frame)
- (3) 主机(H1)向选择的AP发送关联请求帧
- (4) AP向主机(H1)发送关联响应帧





IEEE 802.11:多路访问控制

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

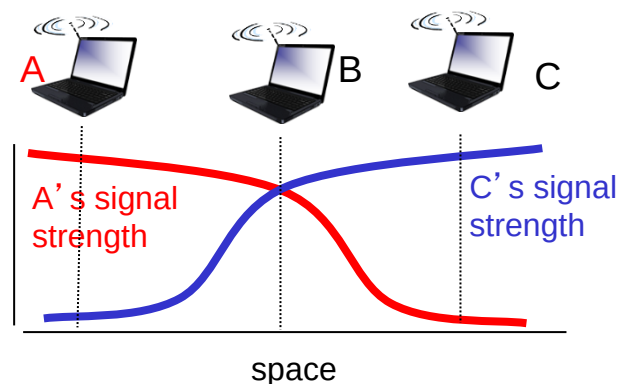
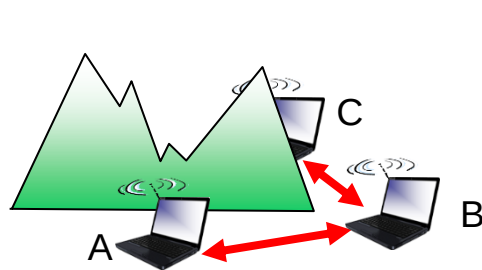
❖ 避免冲突: 2+结点同时传输

❖ 802.11: CSMA – 发送数据前监听信道

- 避免与正在进行传输的其他结点冲突

❖ 802.11: 不能像CSMA/CD那样, 边发送、边检测冲突!

- 无线信道很难实现: 发射信号强度远高于接收信号强度
- 无法侦听到所有可能的冲突: 隐藏站、信号衰落
- 目标: 避免冲突(avoid collisions)-CSMA/C(ollision)A(voidance)





IEEE 802.11 MAC协议: CSMA/CA

7.1 无线与移动网络概述

7.2 无线链路与网络特性

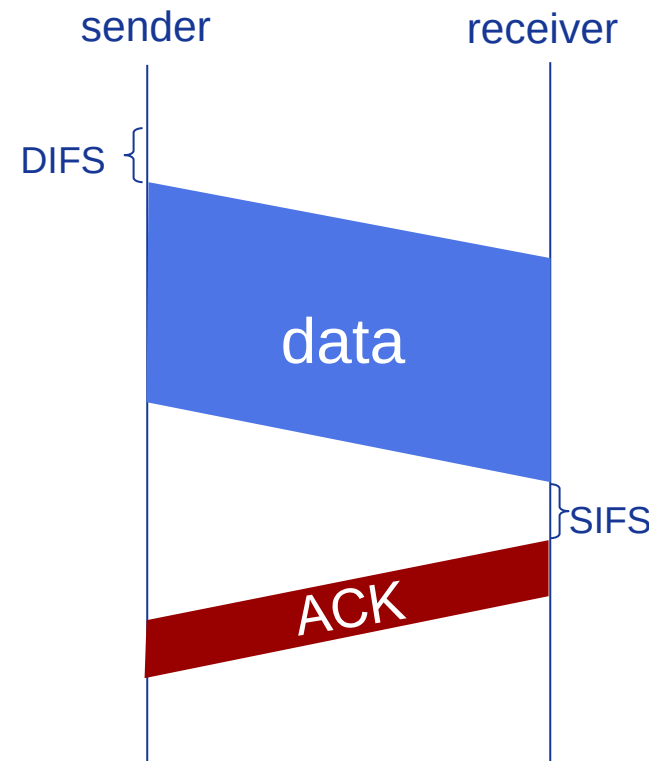
7.3 无线局域网: IEEE802.11

802.11 sender

- 1 if 监听到信道空闲了 **DIFS** 时间 then
发送整个帧(无同时检测冲突, 即CD)
- 2 if 监听到信道忙 then
开始随机退避计时
当信道空闲时, 计时器倒计时
当计时器超时时, 发送帧
if 没有收到ACK then
增加随机退避间隔时间
重复第2步

802.11 receiver

- if 正确接收帧
延迟 **SIFS** 时间后, 向发送端发送ACK
(由于存在隐藏站问题)





IEEE 802.11 MAC协议: CSMA/CA

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网: IEEE802.11

基本思想: 允许发送端“**预约**”(reserve)信道, 而不是随机发送数据帧, 从而避免长数据帧的冲突

- ❖ 发送端首先利用CSMA向BS发送一个很短的**RTS** (request-to-send)帧
 - RTS帧仍然可能彼此冲突 (但RTS帧很短)
- ❖ BS广播一个**CTS**(clear-to-send)帧作为对RTS的响应
- ❖ CTS帧可以被所有结点接收
 - 消除隐藏站影响
 - 发送端可以发送数据帧
 - 其他结点推迟发送

利用很小的预约帧彻底避免了数据帧冲突!



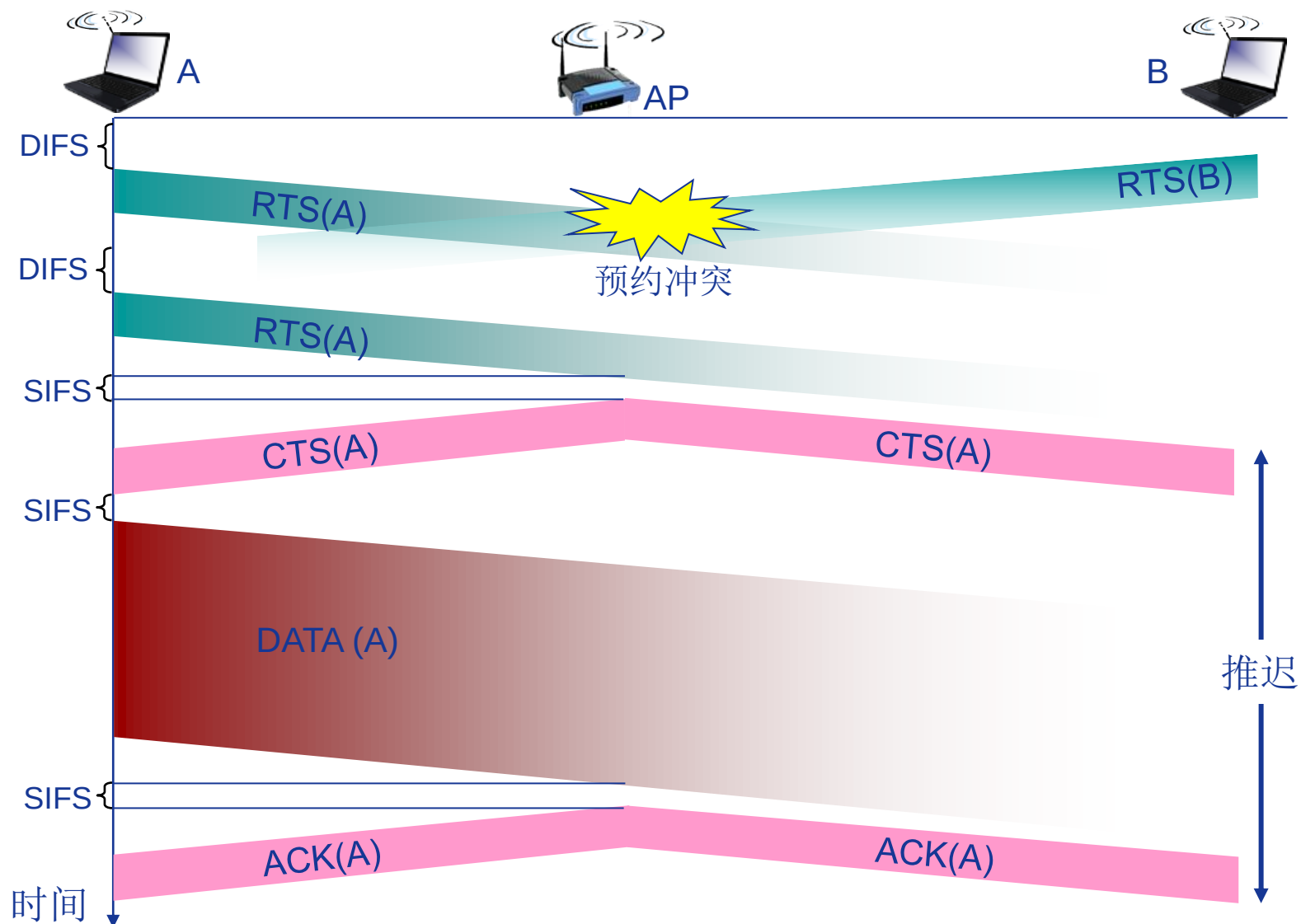


冲突避免(CA): RTS-CTS交换

7.1 无线与移动网络概述

7.2 无线链路与网络特性

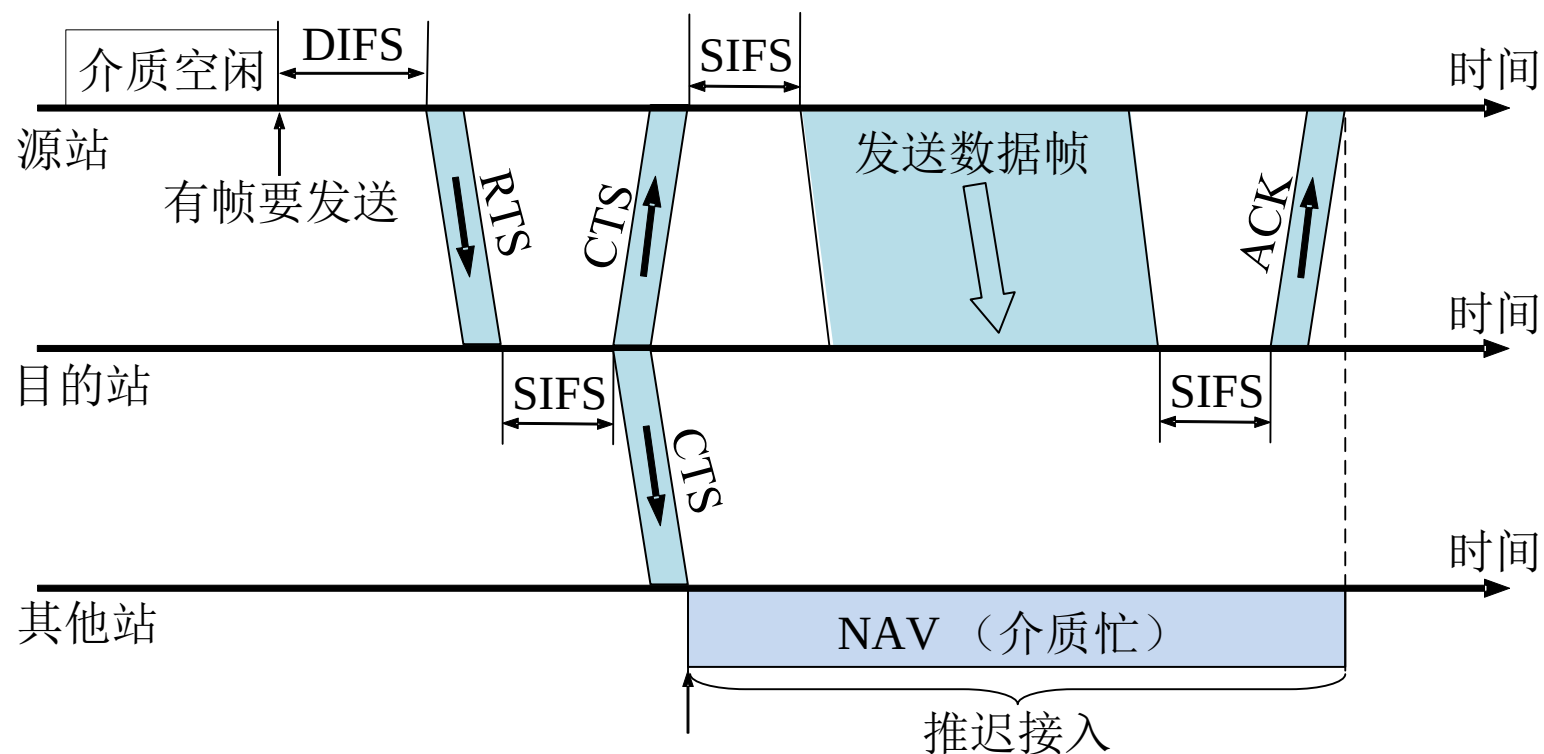
7.3 无线局域网:IEEE802.11



例题

【例1】CSMA/CA协议如何实现信道预约？一个主机从期望发送数据开始，到确认数据被接收方正确接收最快需要多少时间？（注：忽略传播时延、处理时延和传输时延）

DIFS+3SIFS



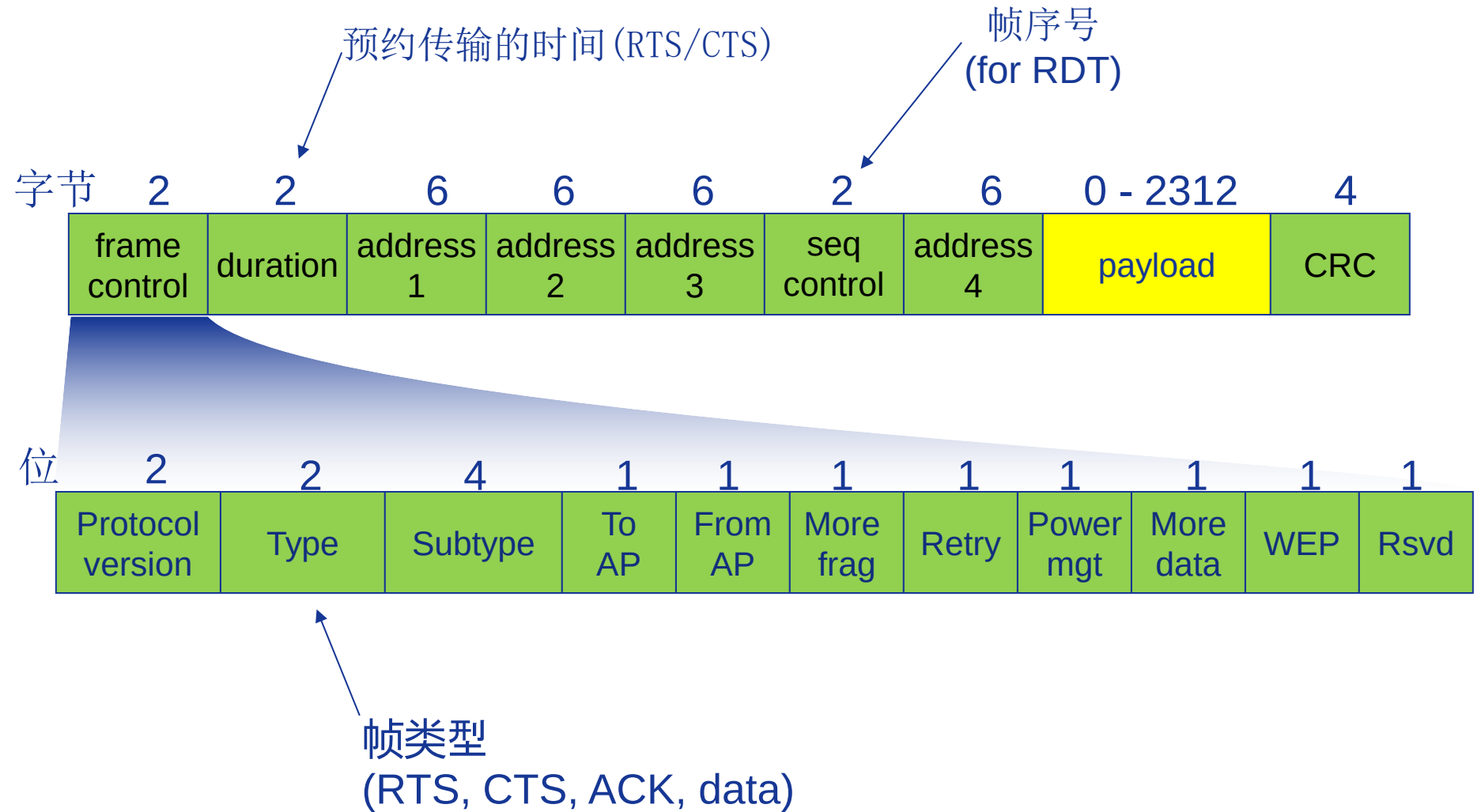


IEEE 802.11 MAC帧

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11





IEEE 802.11数据帧地址

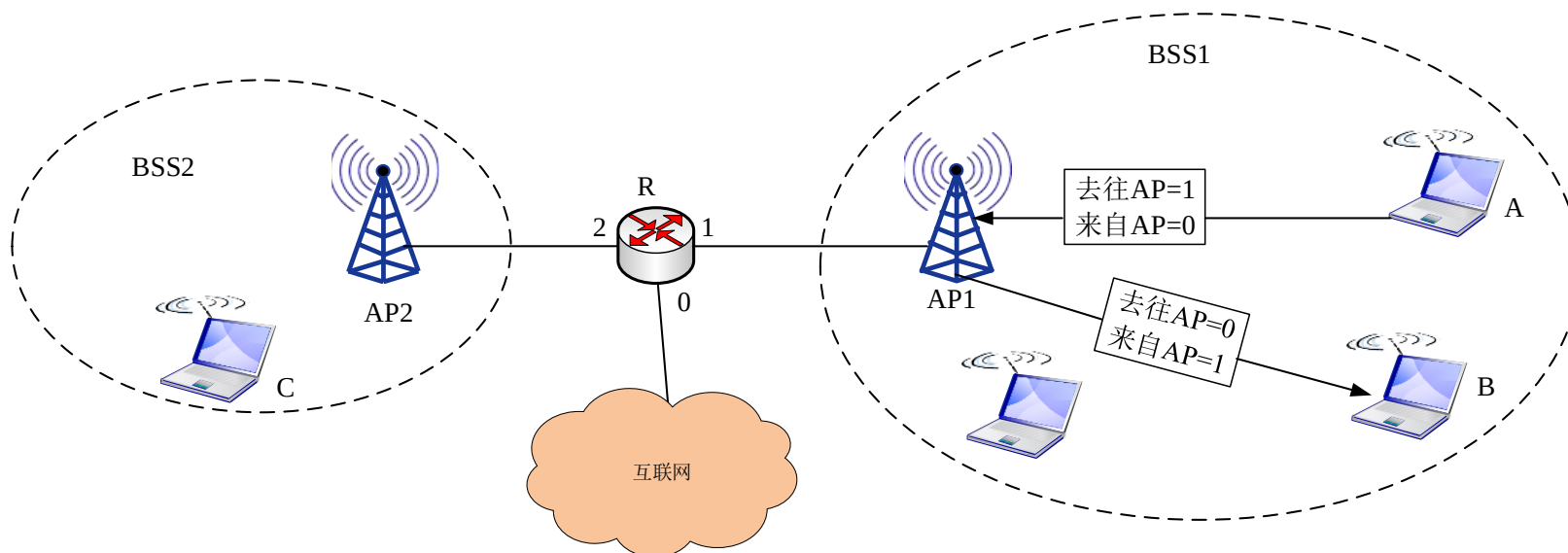
7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

- ❖ 802.11数据帧有4个地址字段
- ❖ 地址 4 用于Ad-hoc或采用无线桥接（无线DS）的情形
- ❖ IEEE802.11帧中的几个地址分别如何取值？

去往AP	来自AP	地址1	地址2	地址3	地址4
0	1	目的地址	AP地址	源地址	——
1	0	AP地址	源地址	目的地址	——



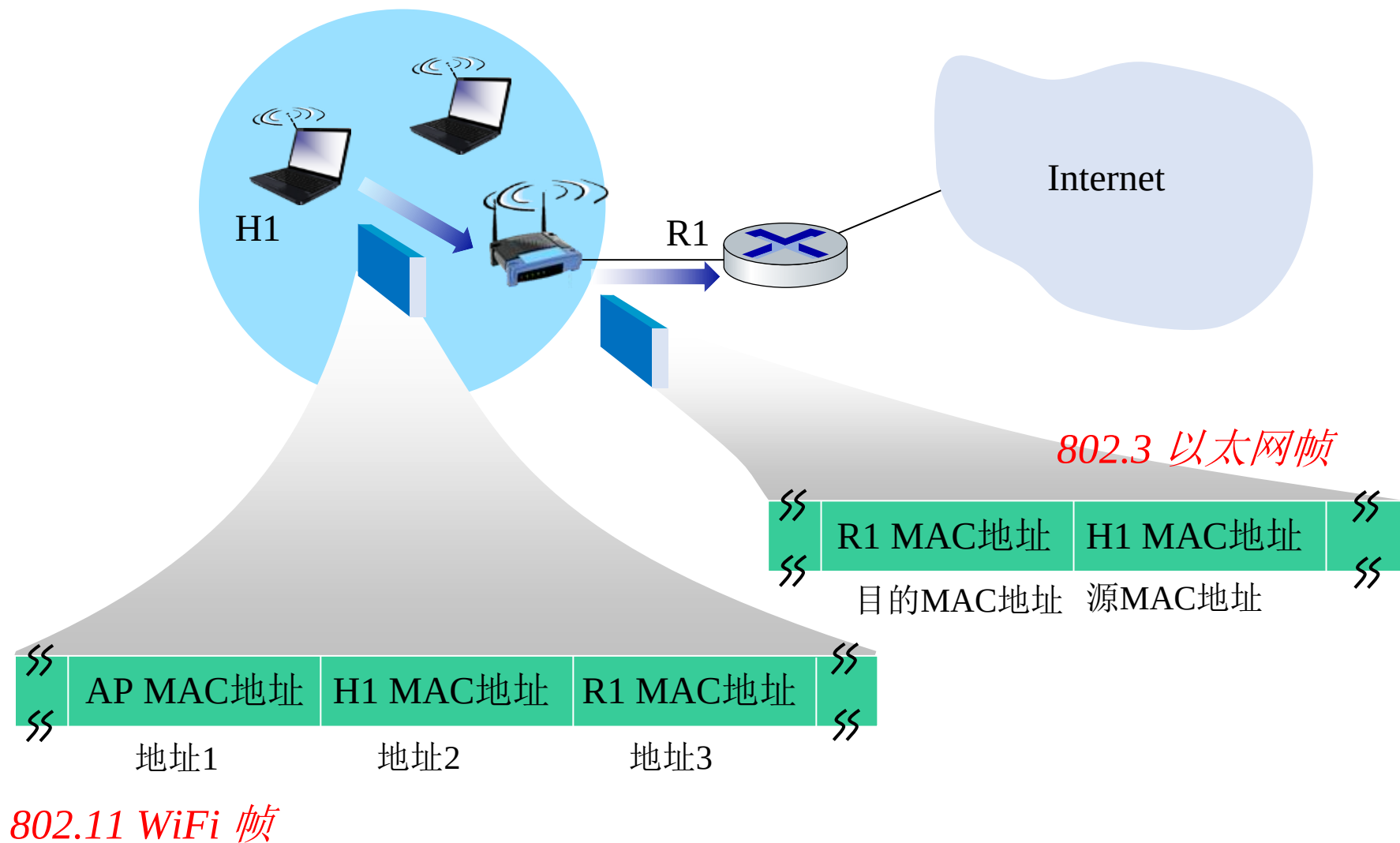


IEEE 802.11帧的地址

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11





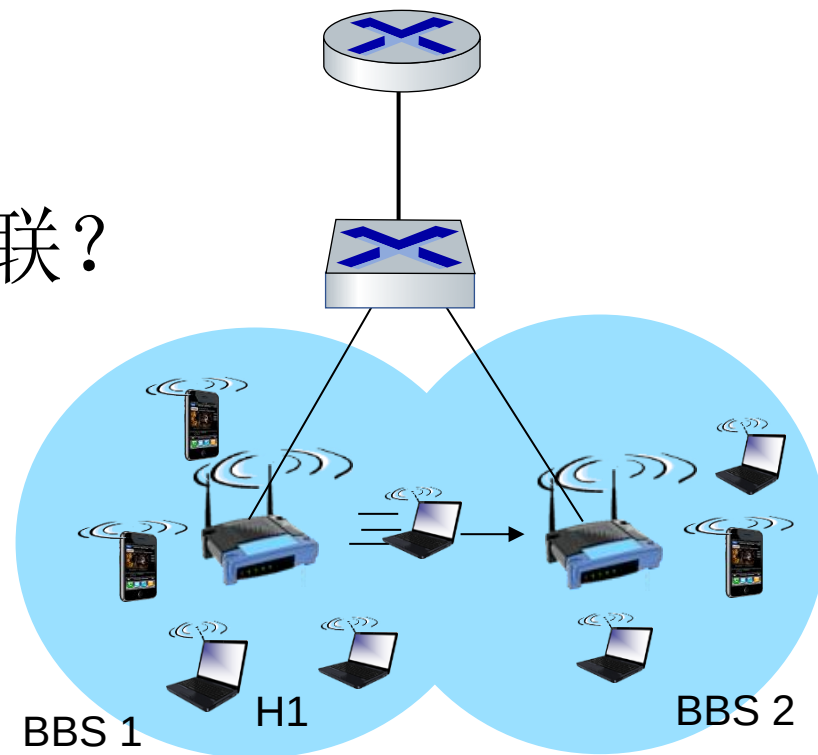
802.11: 同一子网内的移动

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网: IEEE802.11

- H1 留在同一 IP 子网内:
IP 地址可以保持不变
- 交换机: H1与哪个 AP 相关联?
 - 自学习: 交换机根据来自 H1 的帧, “记住” 哪个交换机端口可用到达 H1



802.11：高级功能

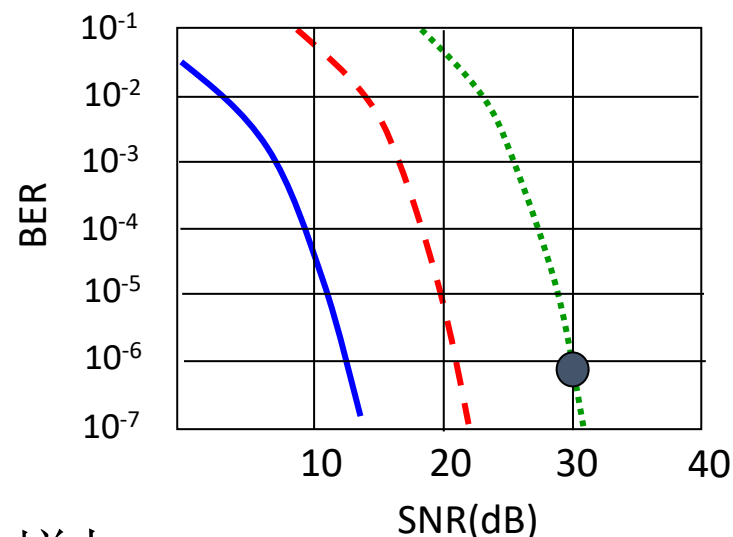
7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

速率适应

- 当移动终端移动、SNR变化时，基站、移动终端动态改变传输速率(通过物理层调制技术)



- 1.当移动终端远离基站时，SNR降低、BER增加
- 2.当 BER 变得过高时，切换到较低传输速率，以降低BER



7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

功率管理

- 结点到AP：“我要休眠，直到下一个信标帧”
 - AP 知道不将帧传输到此结点
 - 结点在下一个信标帧之前醒来
- 信标帧：包含一个**移动终端列表**，AP目前缓存了等待发送给这些移动终端的帧
 - 如果AP到移动终端的帧将要发送，结点将保持唤醒状态；否则，再次休眠，直到下一个信标帧





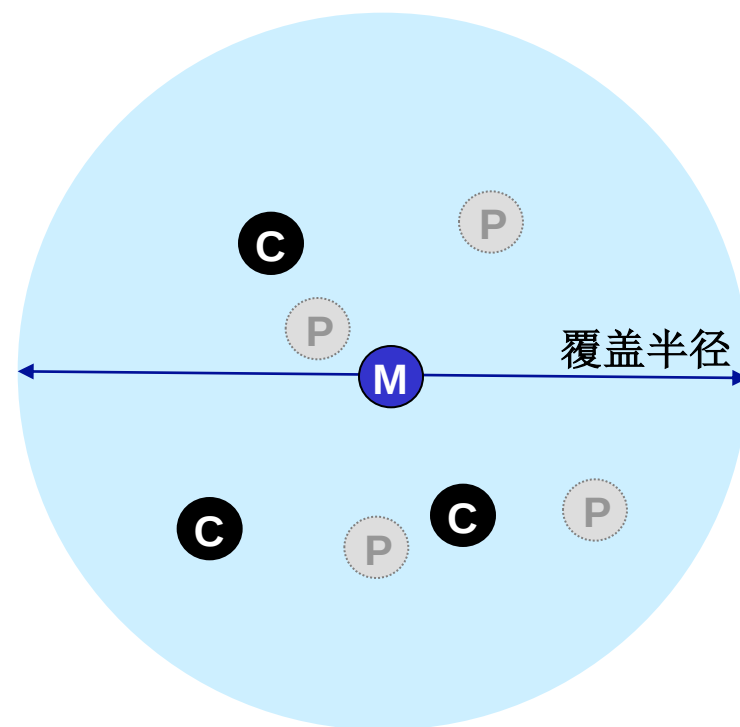
个人区域网: Bluetooth

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网: IEEE802.11

- 直径小于 10 m
- 替代线缆（鼠标、键盘、耳机）
- 自组网：无基础设施
- 2.4-2.5 GHz ISM 无线电频段，高达 4 Mbps
- 主控制器/客户设备：
 - 主控制器轮询客户端，授予客户传输请求



- M 主设备(master device)
- C 客户设备(client device)
- P 停放设备(parked device (inactive))





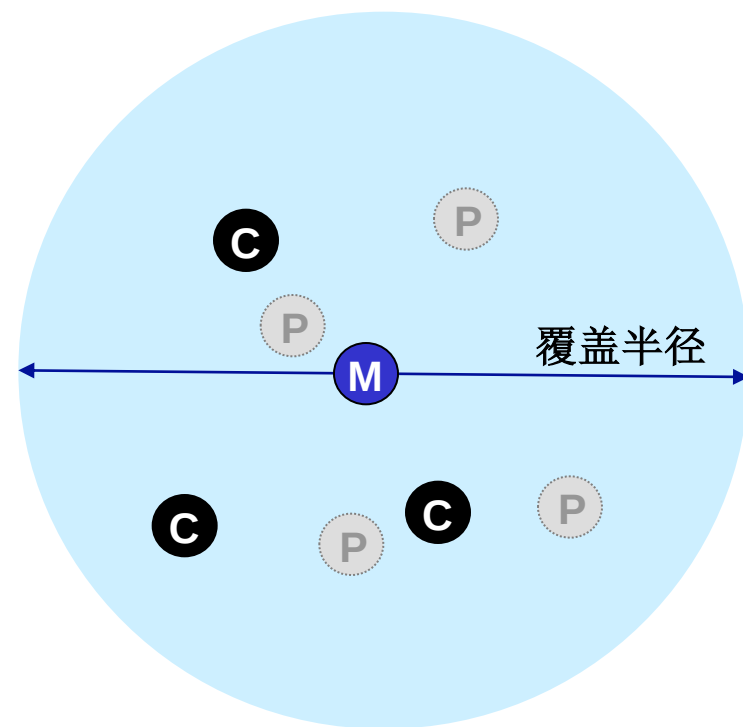
个人区域网: Bluetooth

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网: IEEE802.11

- TDM: 625 μ sec时隙
- FDM: 发送方使用跳频扩频 (FHSS)
 - 79个频率信道, 每个时隙使用一个信道
 - 时隙到时隙, 按伪随机顺序在79个信道间变更信道
 - 其他不在皮克网 (piconet) 的设备仅可能干扰某些时隙
- 停放模式: 客户设备可以“进入睡眠” (停放) 和稍后唤醒 (以节省电池)
- 引导 (bootstrapping): 节点自组 (即插即用) 成皮克网



- M 主设备 (master device)
- C 客户设备 (client device)
- P 停放设备 (parked device (inactive))





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7.4 移动通信网络体系结构

聂兰顺



Components of cellular network architecture

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

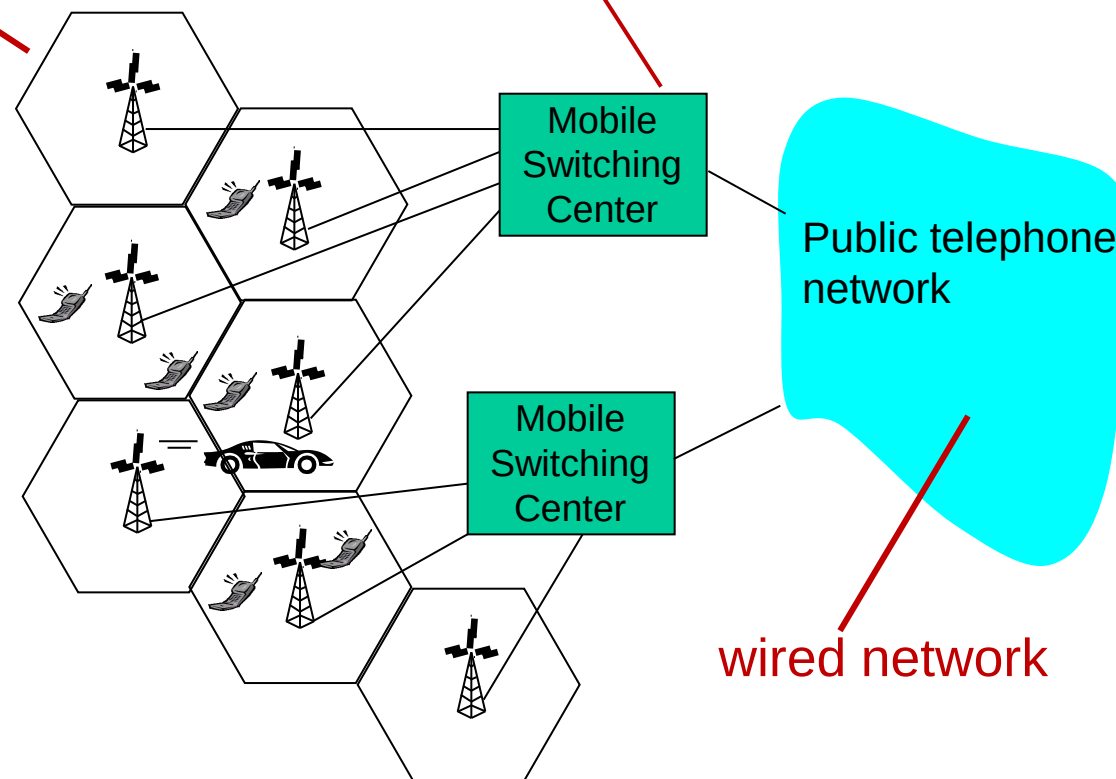
7.4 移动通信网络体系结构

cell

- ❖ covers geographical region
- ❖ *base station* (BS)
analogous to 802.11 AP
- ❖ *mobile users* attach to network through BS
- ❖ *air-interface*: physical and link layer protocol between mobile and BS

MSC

- ❖ connects cells to wired tel. net.
- ❖ manages call setup (more later!)
- ❖ handles mobility (more later!)





Cellular networks: the first hop

7.1 无线与移动网络概述

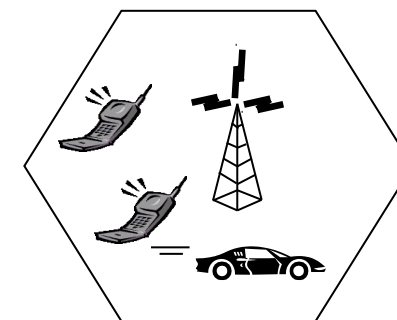
7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

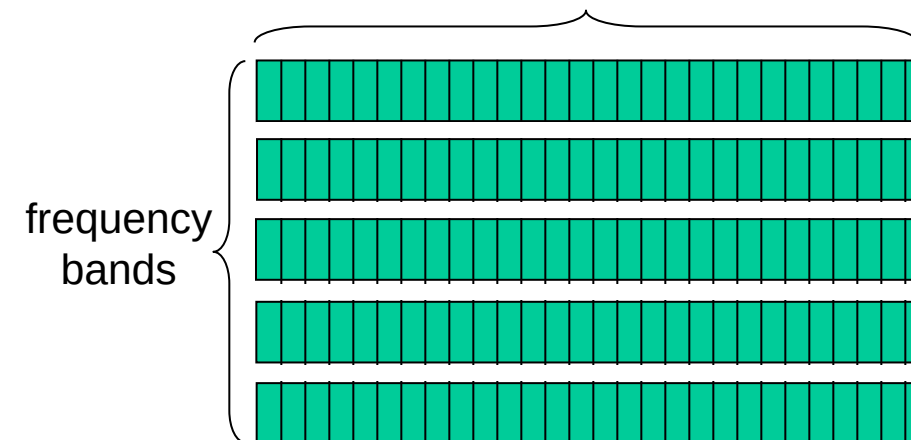
7.4 移动通信网络体系结构

Two techniques for sharing
mobile-to-BS radio spectrum

- ❖ **combined FDMA/TDMA:**
divide spectrum in frequency
channels, divide each channel
into time slots
- ❖ **CDMA:** code division multiple
access



time slots





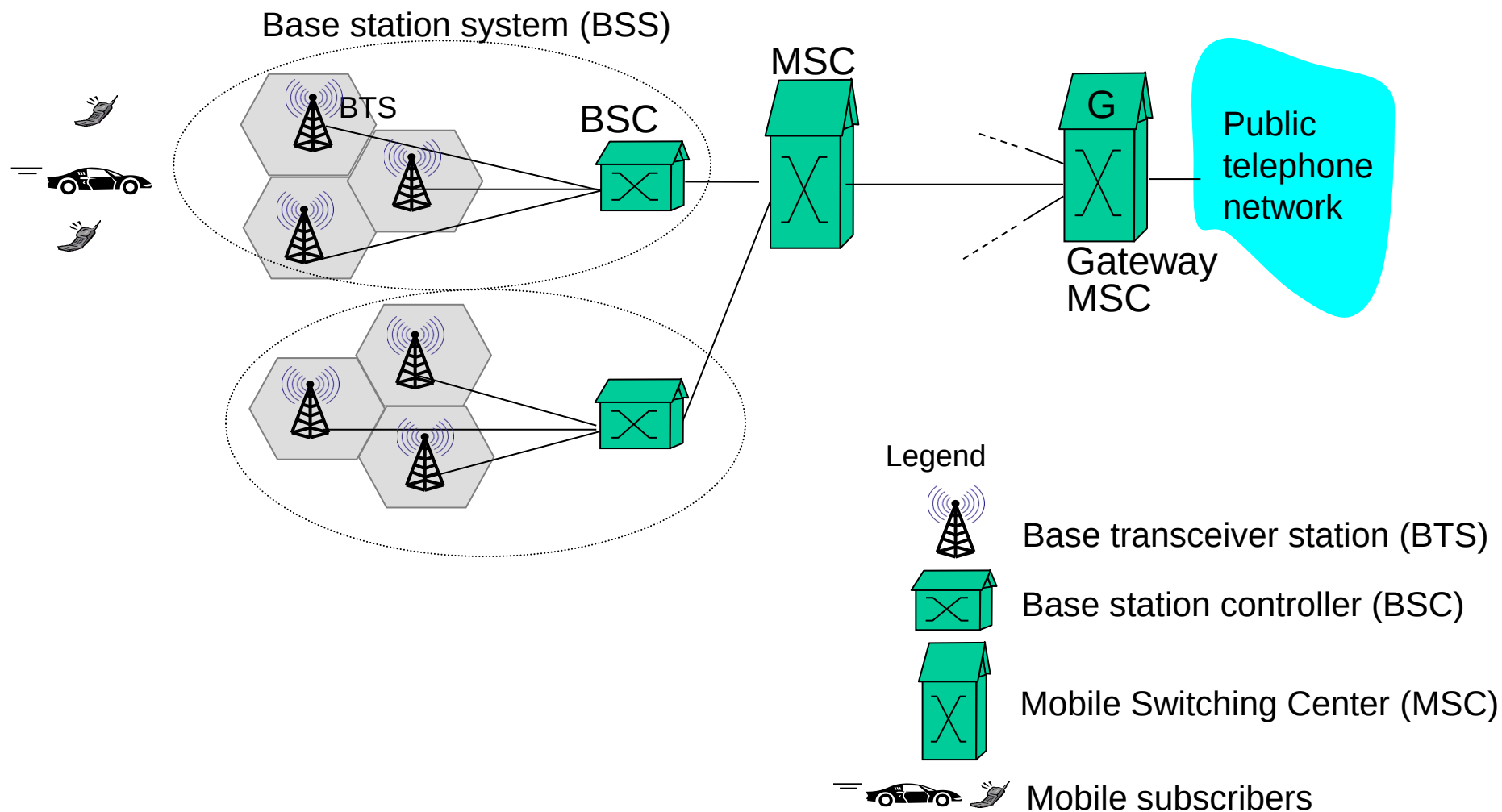
2G (voice) network architecture

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构





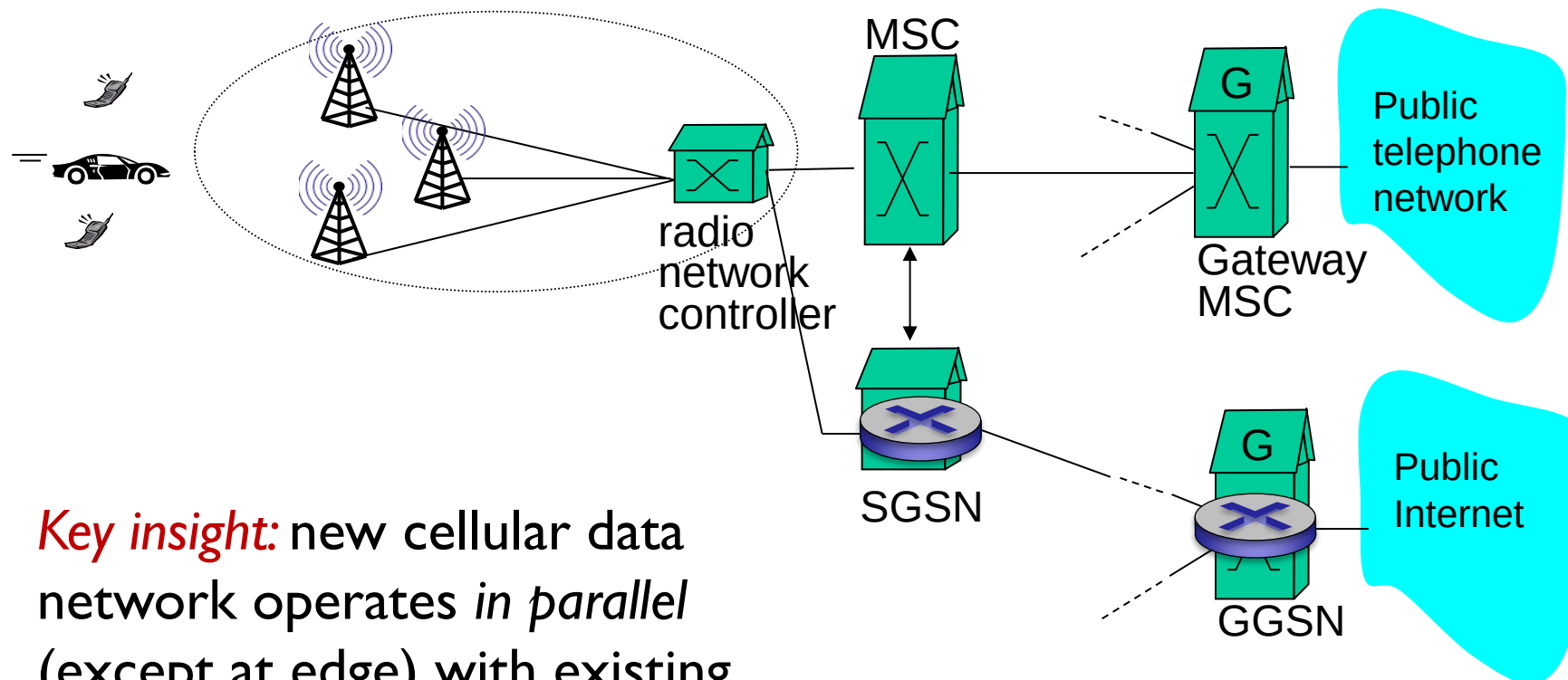
3G (voice+data) network architecture

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7.2 无线链路与网络特性

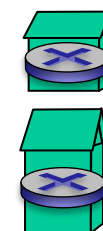
7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构



Key insight: new cellular data network operates *in parallel* (except at edge) with existing cellular voice network

- voice network **unchanged** in core
- data network operates in parallel



Serving GPRS Support Node (SGSN)

Gateway GPRS Support Node (GGSN)





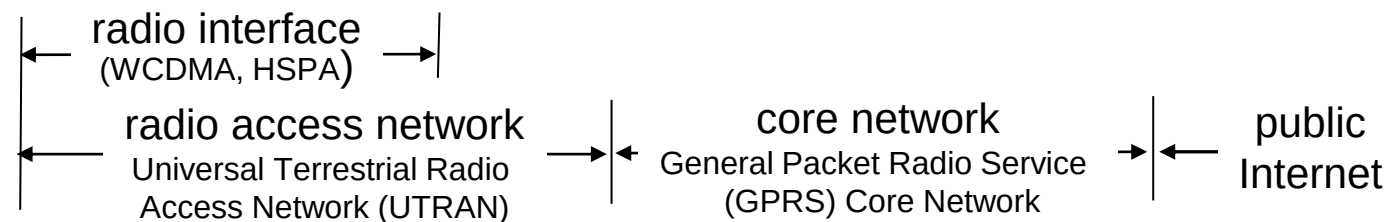
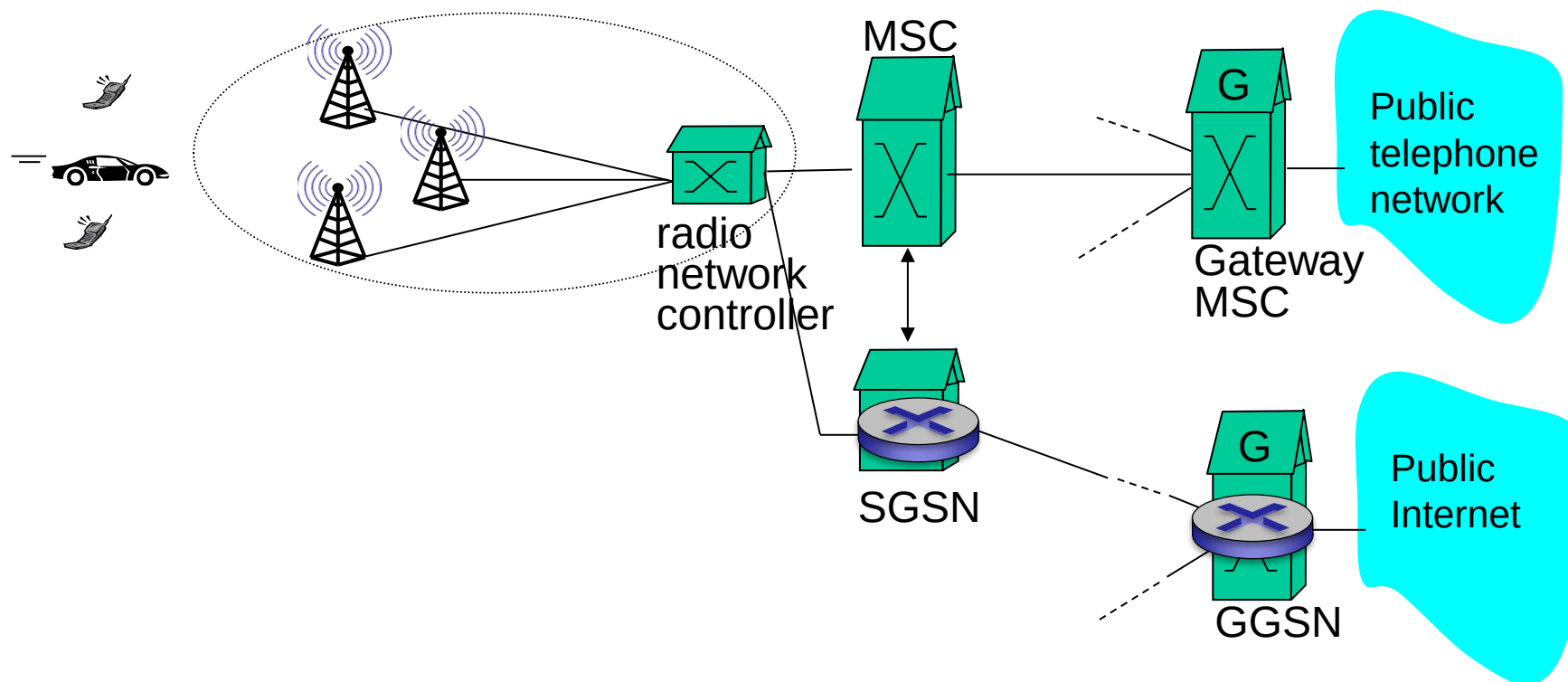
3G (voice+data) network architecture

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7.4 移动通信网络体系结构



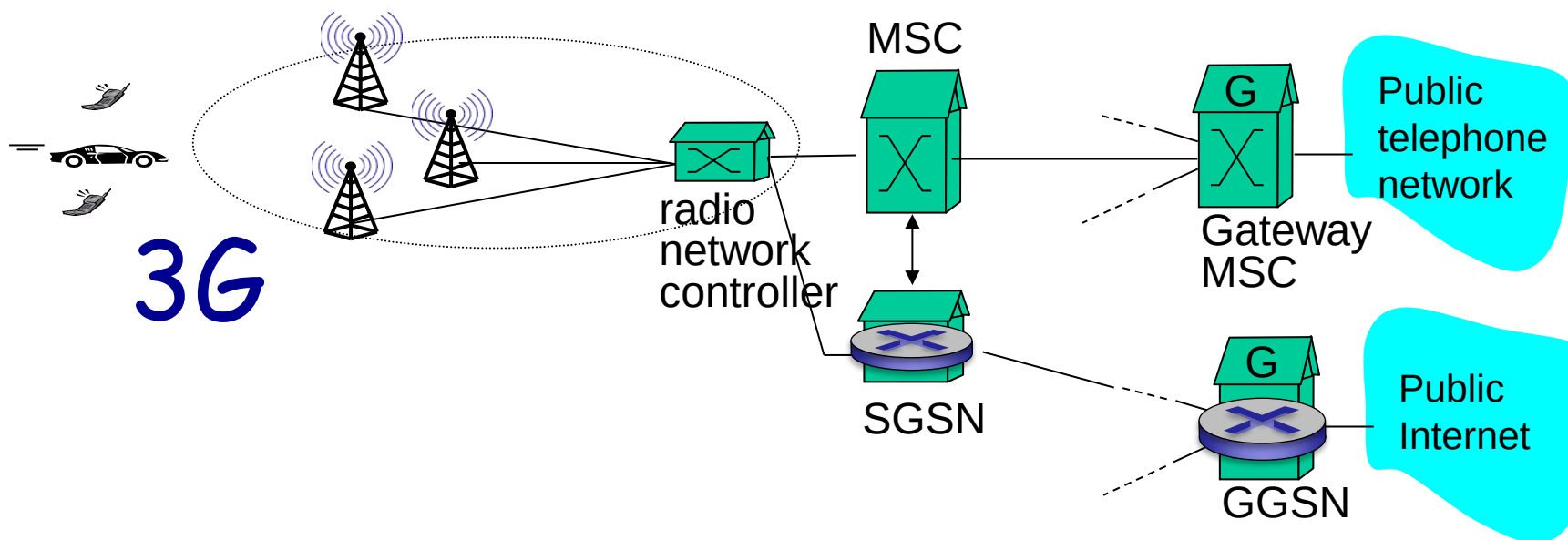
3G versus 4G LTE network architecture

7.1 无线与移动网络概述

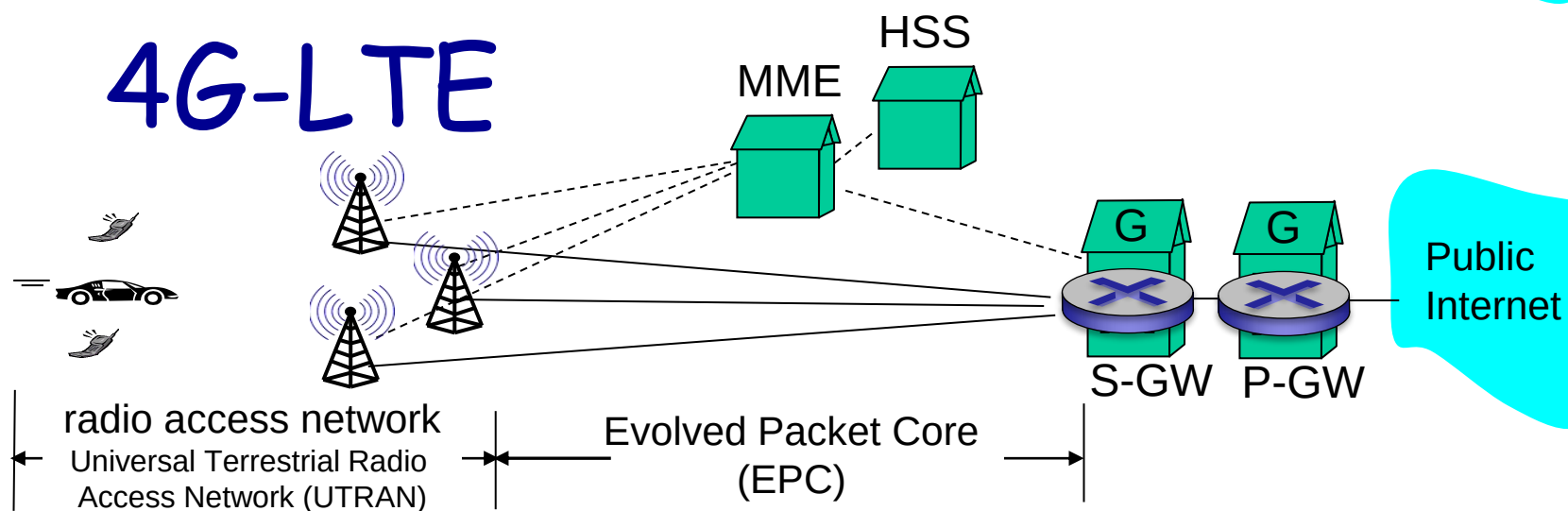
7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

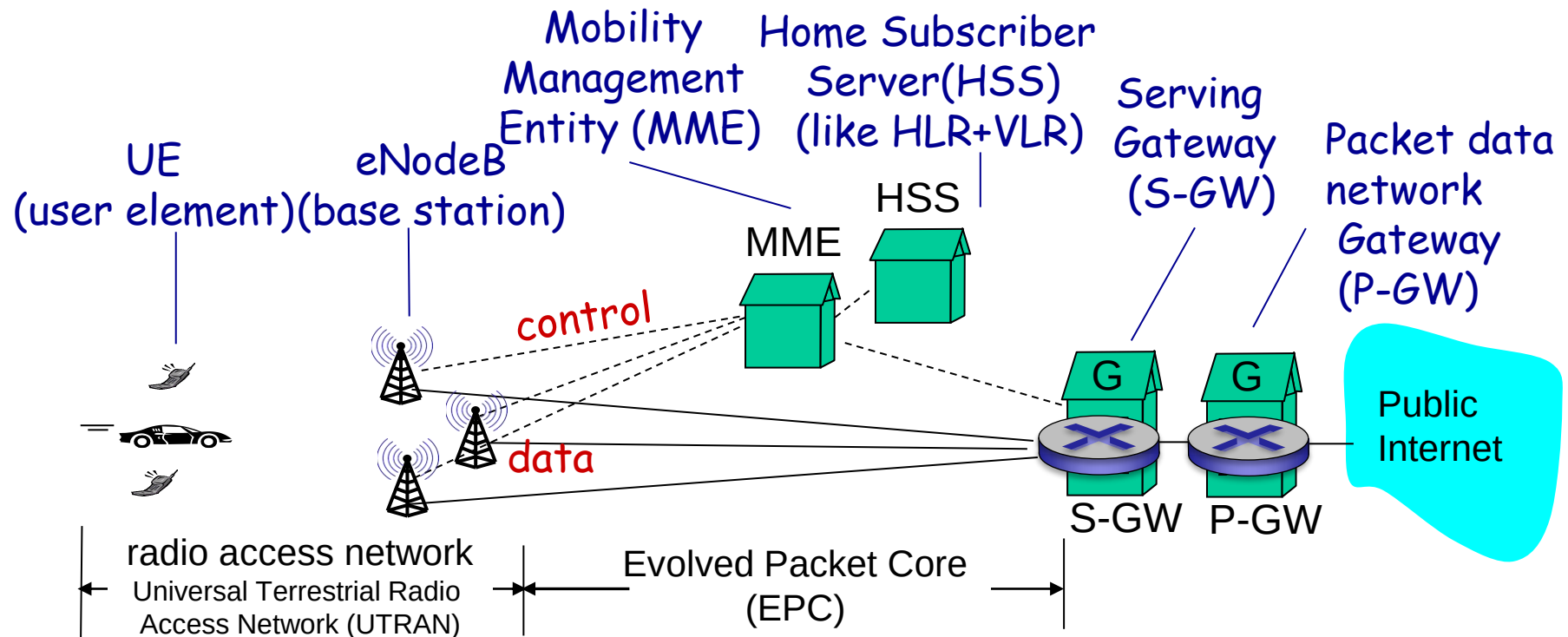


4G-LTE



4G: differences from 3G

- ❖ all IP core: IP packets tunneled (through core IP network) from base station to gateway
- ❖ no separation between voice and data – all traffic carried over IP core to gateway





4G/5G cellular networks

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

- *the* solution for wide-area mobile Internet
- widespread deployment/use:
 - more mobile-broadband-connected devices than fixed-broadband-connected devices (5-1 in 2019)!
 - 4G availability: 97% of time in Korea (90% in US)
- transmission rates up to 100's Mbps
- technical standards: 3rd Generation Partnership Project (3GPP)
 - www.3gpp.org
 - 4G: Long-Term Evolution (LTE) standard





4G/5G cellular networks

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

❖ *similarities to wired Internet*

- edge/core distinction, but both below to same carrier
- global cellular network: a network of networks
- widespread use of protocols we've studied: HTTP, DNS, TCP, UDP, IP, NAT, separation of data/control planes, SDN, Ethernet, tunneling
- interconnected to wired Internet

❖ *differences from wired Internet*

- different wireless link layer
- mobility as a 1st class service
- user “identity” (via SIM card)
- business model: users subscribe to a cellular provider
 - strong notion of “home network” versus roaming on visited nets
 - global access, with authentication infrastructure, and inter-carrier settlements



Elements of 4G LTE architecture

7.1 无线与移动网络概述

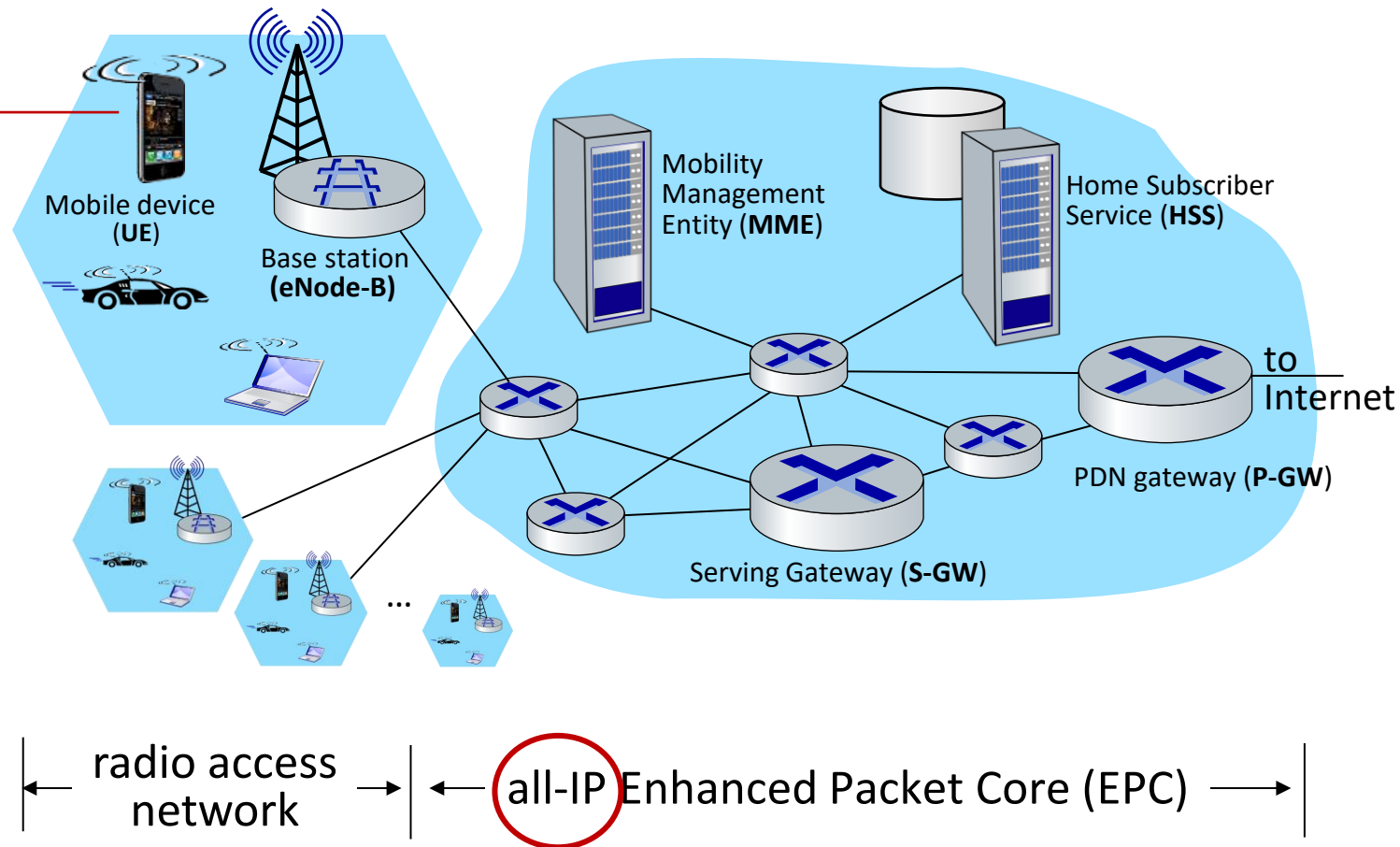
7.2 无线链路和网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

Mobile device:

- smartphone, tablet, laptop, IoT, ... with 4G LTE radio
- 64-bit International Mobile Subscriber Identity (IMSI), stored on SIM (Subscriber Identity Module) card
- LTE jargon: User Equipment (UE)





Elements of 4G LTE architecture

7.1 无线与移动网络概述

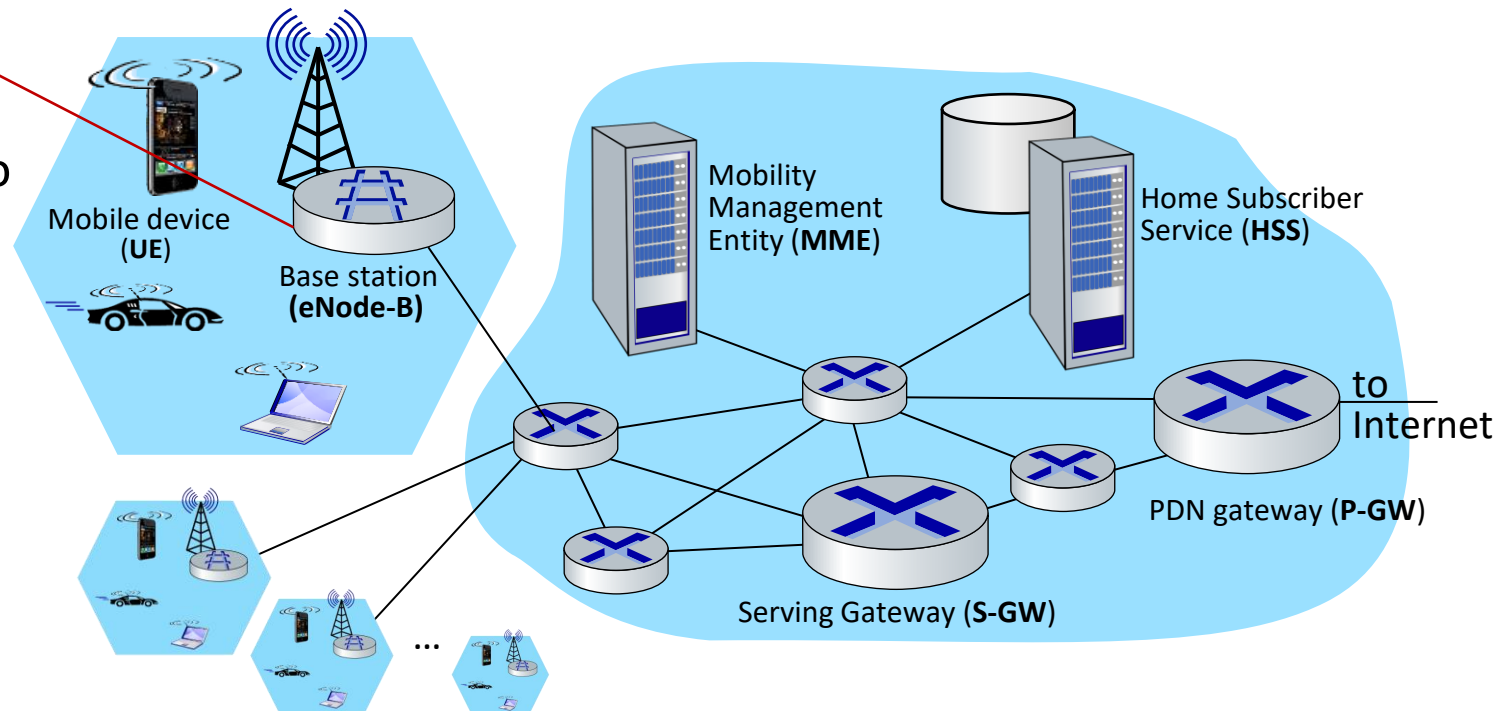
7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

Base station:

- at “edge” of carrier’s network
- manages wireless radio resources, mobile devices in its coverage area (“cell”)
- coordinates device authentication with other elements
- similar to WiFi AP but:
 - active role in user mobility
 - coordinates with nearly base stations to optimize radio use
- LTE jargon: eNode-B





Elements of 4G LTE architecture

7.1 无线与移动网络概述

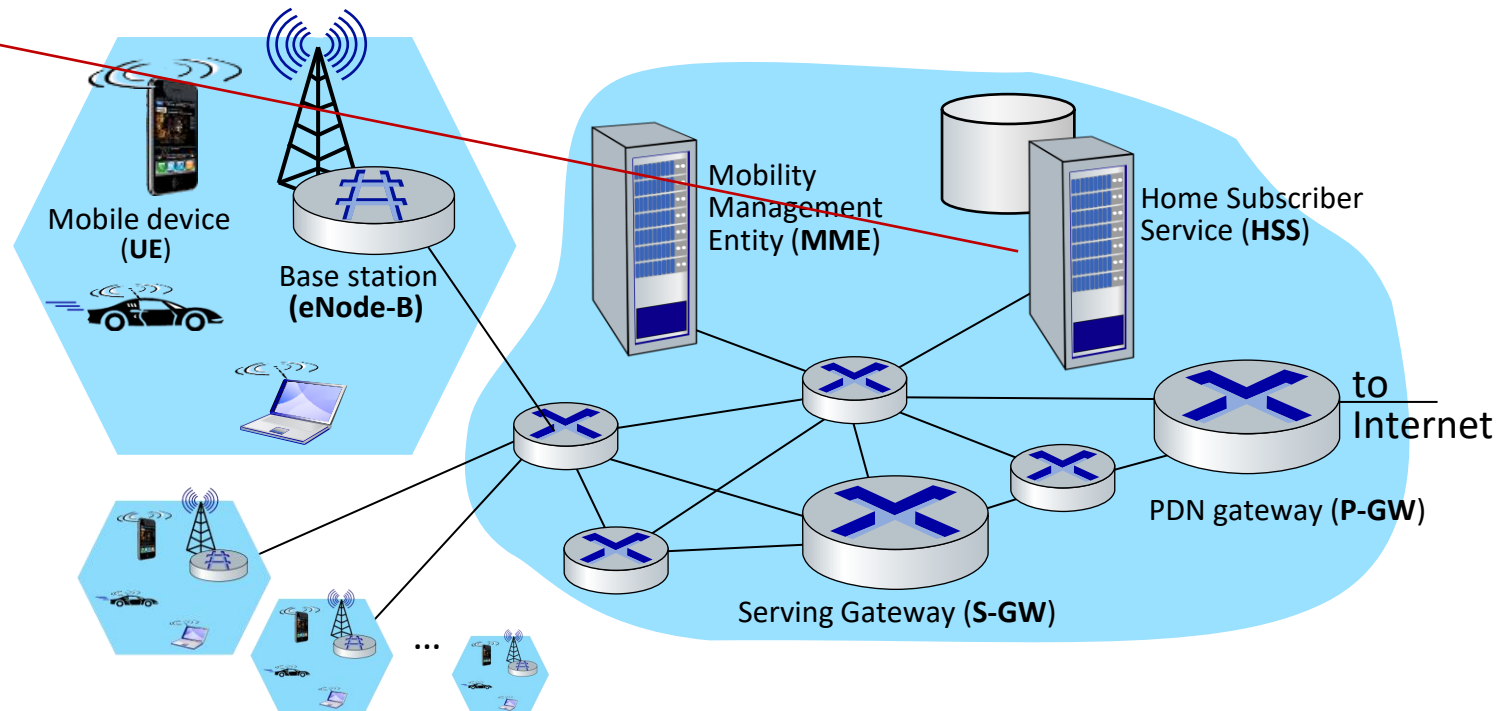
7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

Home Subscriber Service

- stores info about mobile devices for which the HSS's network is their "home network"
- works with MME in device authentication



Elements of 4G LTE architecture

7.1 无线与移动网络概述

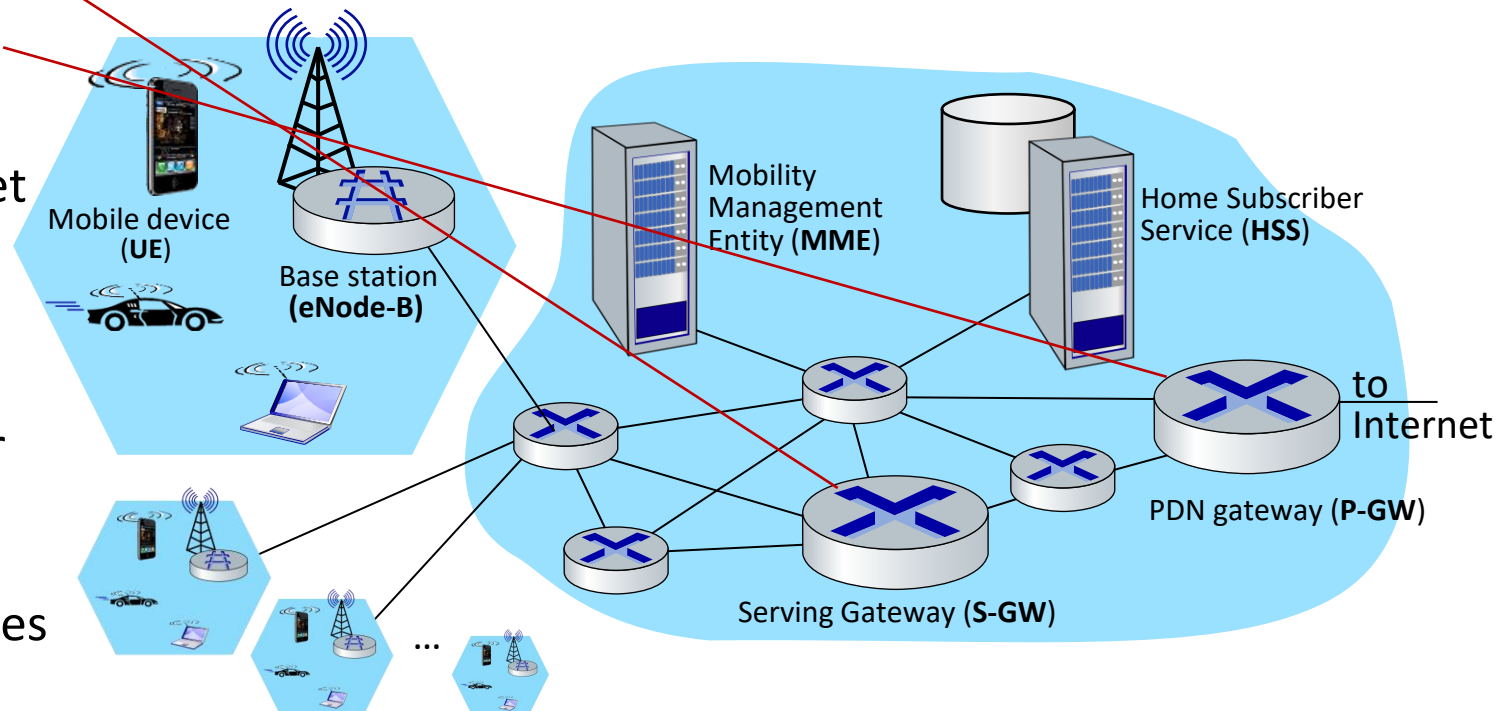
7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

Serving Gateway (S-GW), PDN Gateway (P-GW)

- lie on data path from mobile to/from Internet
- P-GW
 - gateway to mobile cellular network
 - Looks like any other internet gateway router
 - provides NAT services
- other routers:
 - extensive use of tunneling





Elements of 4G LTE architecture

7.1 无线与移动网络概述

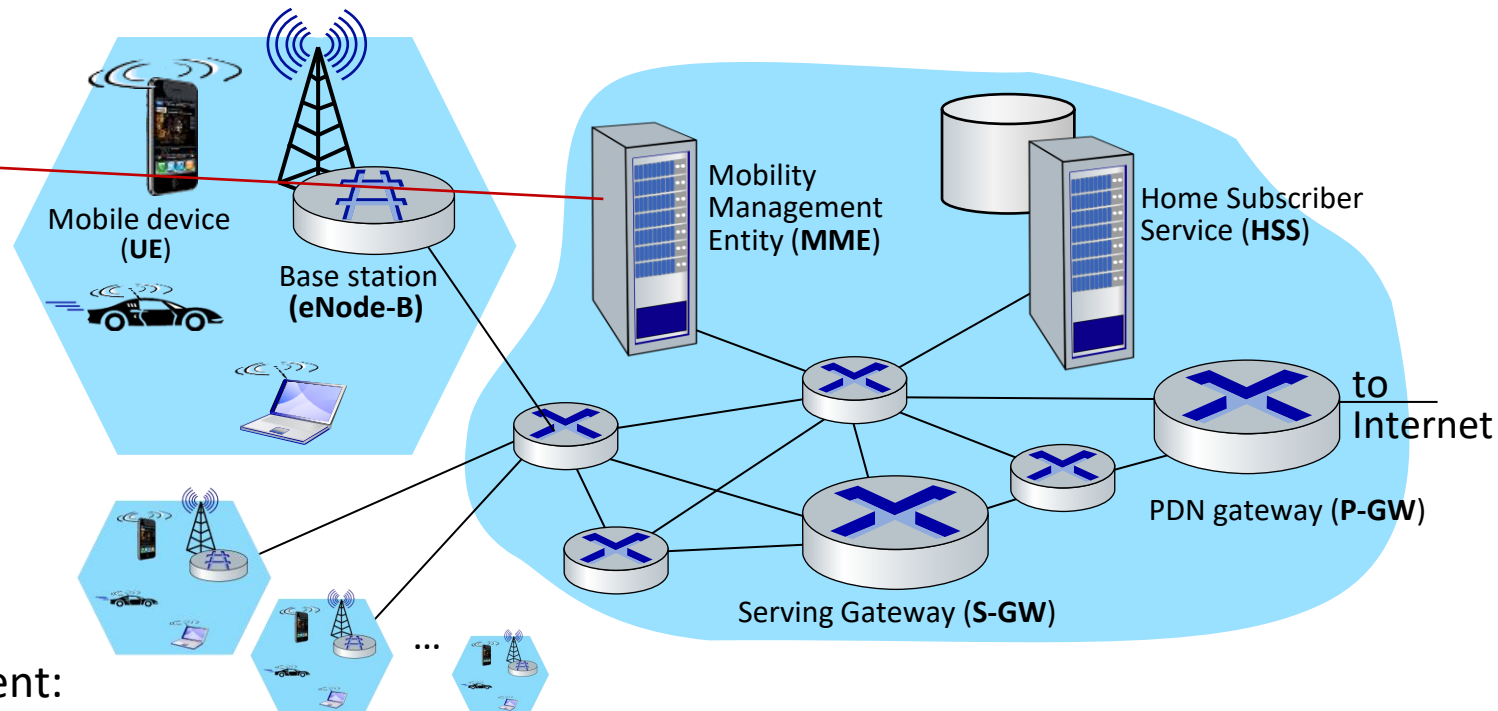
7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

Mobility Management Entity

- device authentication (device-to-network, network-to-device) coordinated with mobile home network HSS
- mobile device management:
 - device handover between cells
 - tracking/paging device location
- path (tunneling) setup from mobile device to P-GW





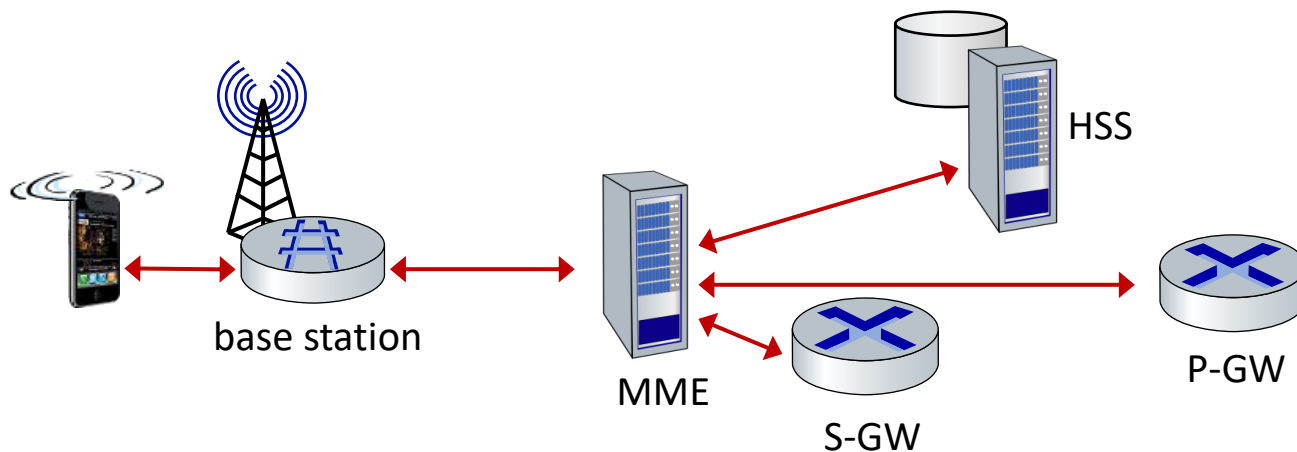
LTE: data plane control plane separation

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7.2 无线链路与网络特性

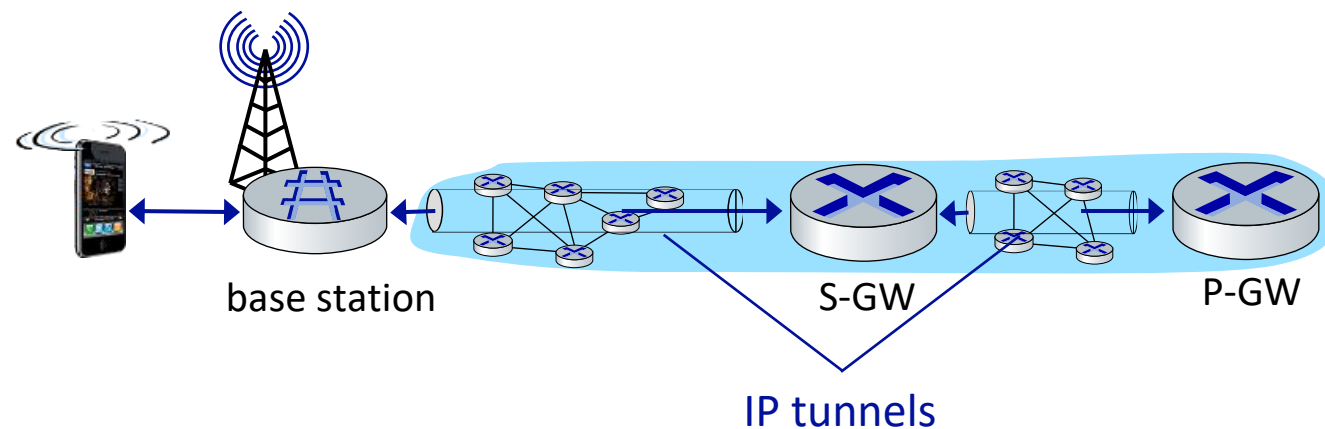
7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构



control plane

- new protocols for mobility management, security, authentication (later)



data plane

- new protocols at link, physical layers
- extensive use of tunneling to facilitate mobility





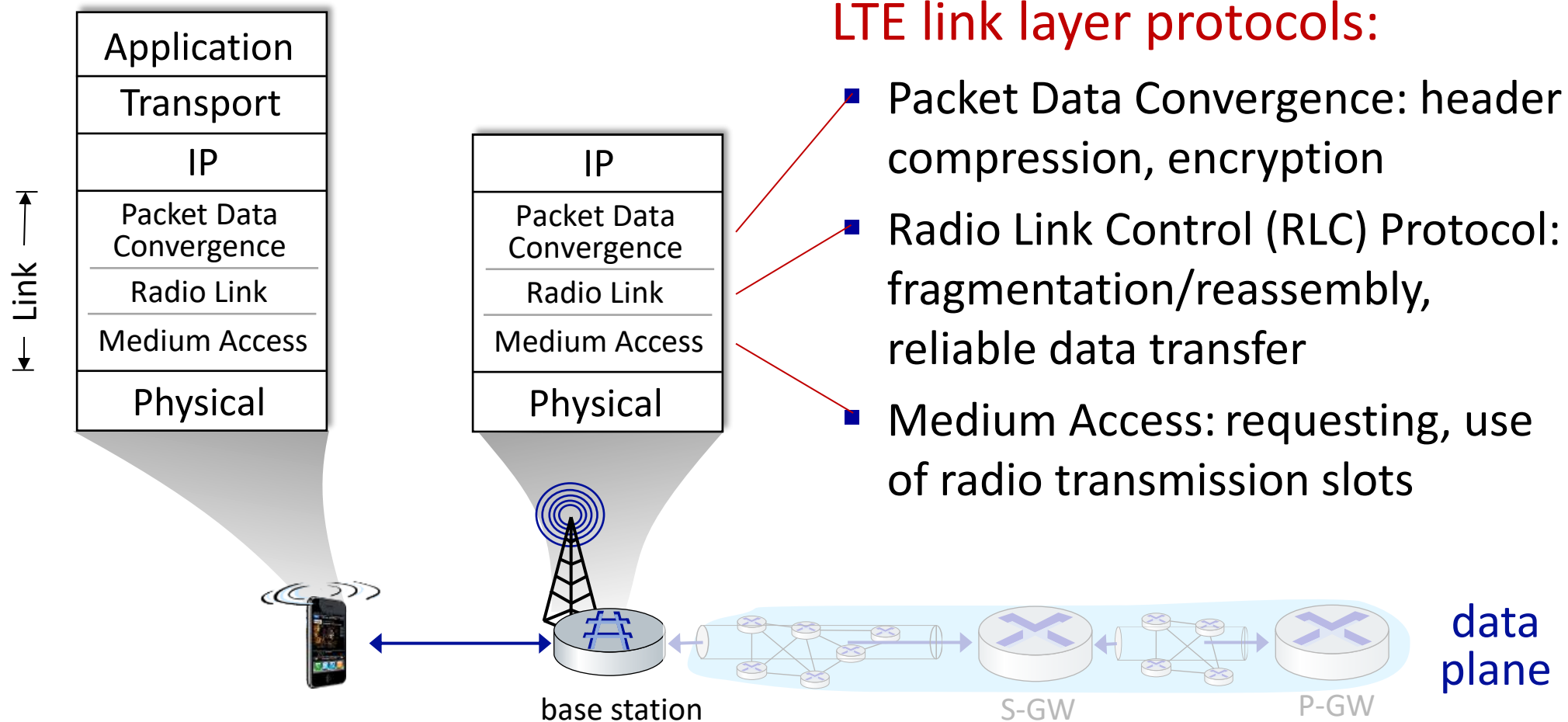
LTE data plane protocol stack: first hop

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构





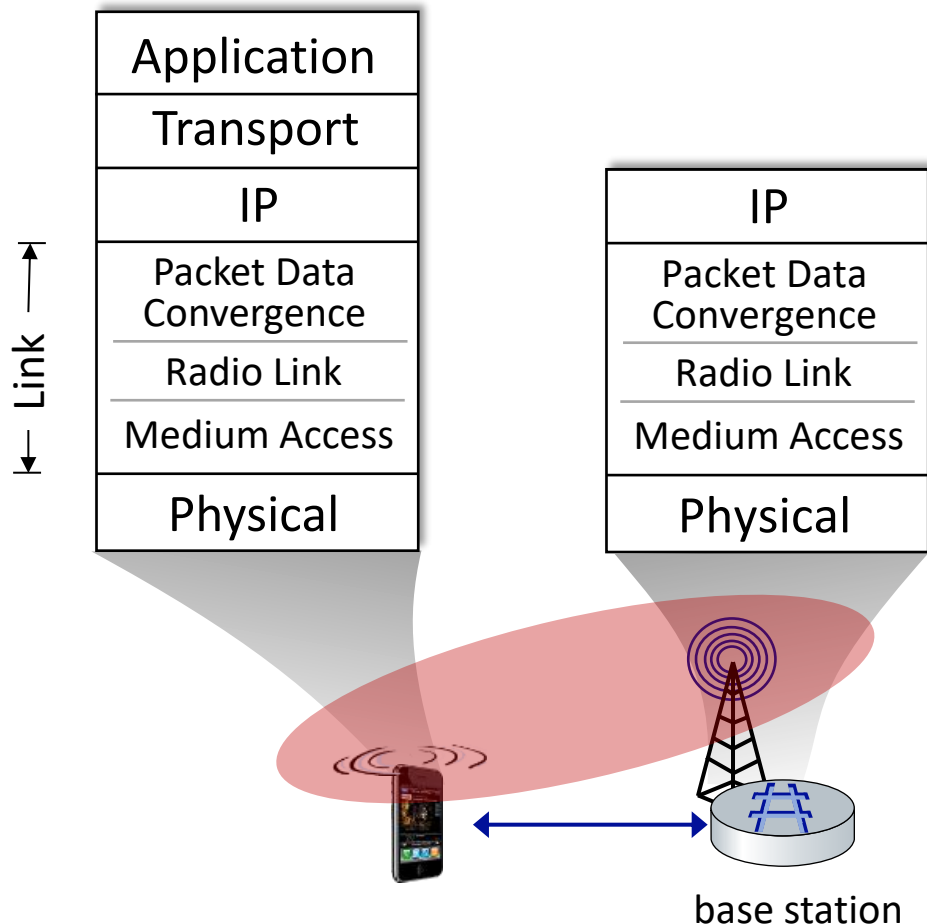
LTE data plane protocol stack: first hop

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构



LTE radio access network:

- **downstream channel:** FDM, TDM within frequency channel (OFDM - orthogonal frequency division multiplexing)
 - “orthogonal”: minimal interference between channels
 - **upstream:** FDM, TDM similar to OFDM
- each active mobile device allocated two or more 0.5 ms time slots over 12 frequencies
 - scheduling algorithm not standardized – up to operator
 - 100's Mbps per device possible



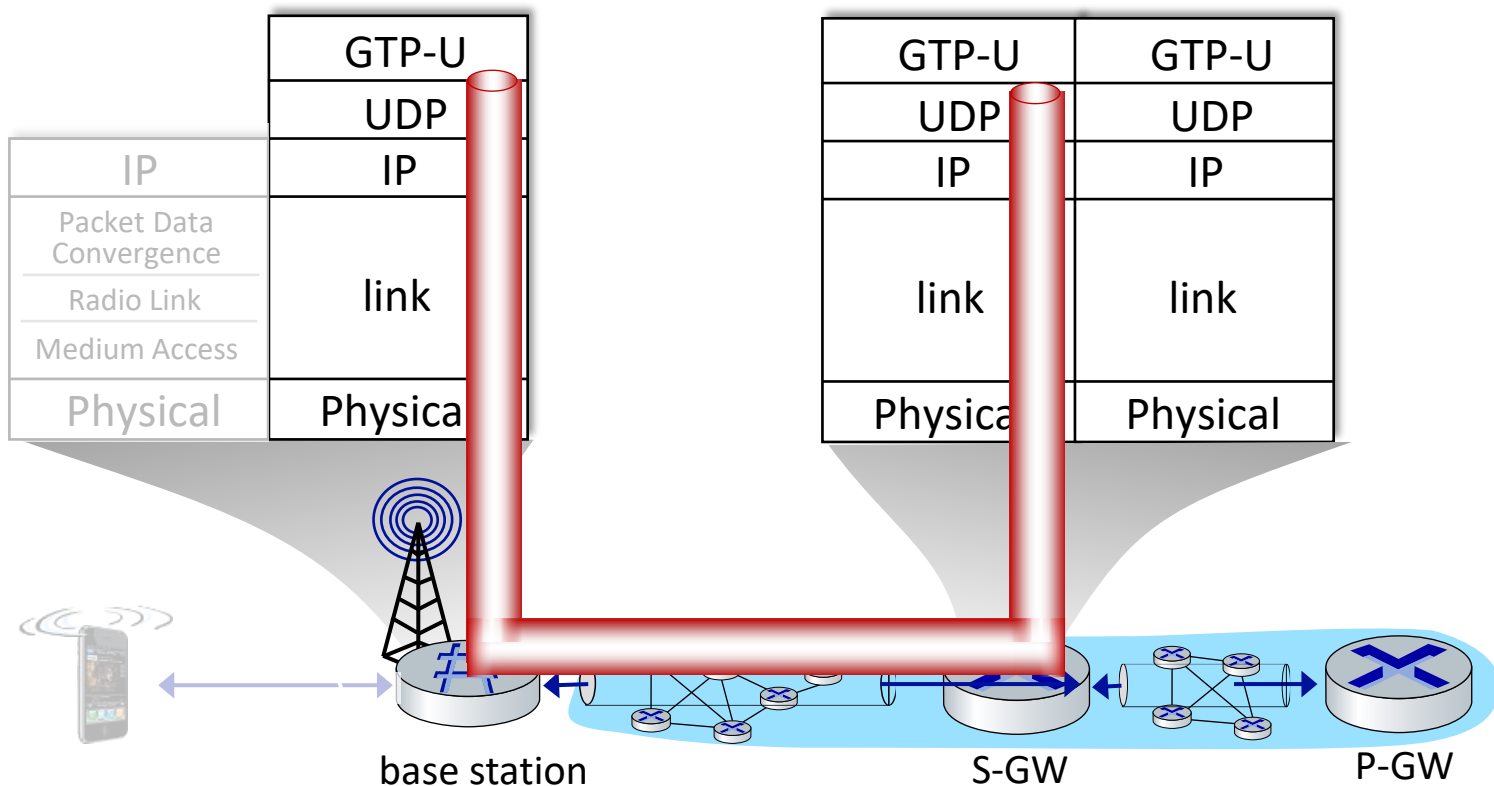
LTE data plane protocol stack: packet core

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构



tunneling:

- mobile datagram encapsulated using GPRS Tunneling Protocol (GTP), sent inside UDP datagram to S-GW
- S-GW re-tunnels datagrams to P-GW
- supporting mobility: only tunneling endpoints change when mobile user moves





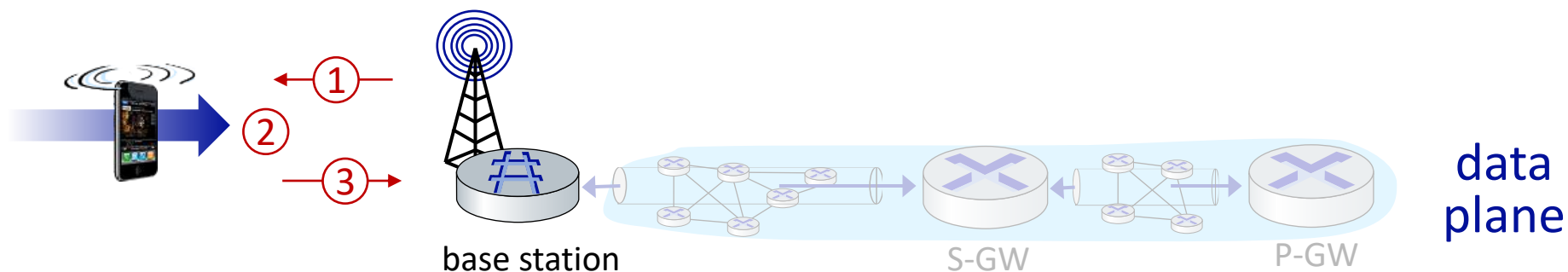
LTE data plane: associating with a BS

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构



- ① BS broadcasts primary synch signal every 5 ms on all frequencies
 - BSs from multiple carriers may be broadcasting synch signals
- ② mobile finds a primary synch signal, then locates 2nd synch signal on this freq.
 - mobile then finds info broadcast by BS: channel bandwidth, configurations; BS's cellular carrier info
 - mobile may get info from multiple base stations, multiple cellular networks
- ③ mobile selects which BS to associate with (*e.g.*, preference for home carrier)
- ④ more steps still needed to authenticate, establish state, set up data plane





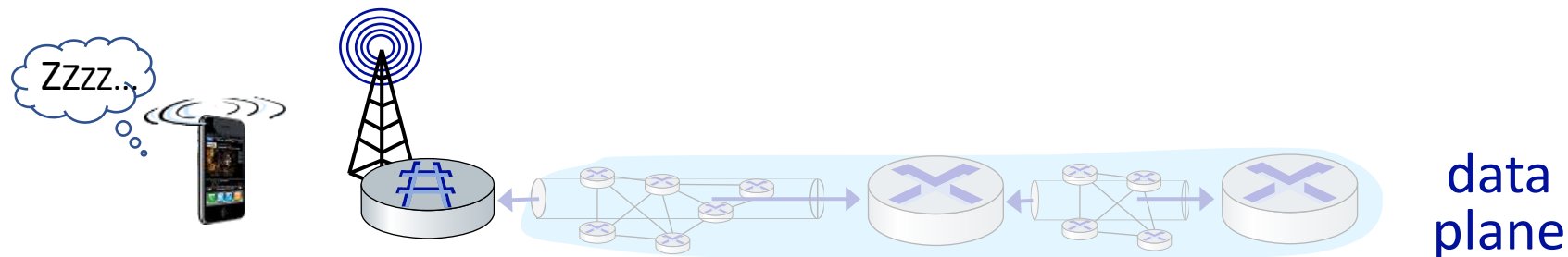
LTE mobiles: sleep modes

7.1 无线与移动网络概述

7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构



as in WiFi, Bluetooth: LTE mobile may put radio to “sleep” to conserve battery:

- **light sleep**: after 100's msec of inactivity
 - wake up periodically (100's msec) to check for downstream transmissions
- **deep sleep**: after 5-10 secs of inactivity
 - mobile may change cells while deep sleeping – need to re-establish association





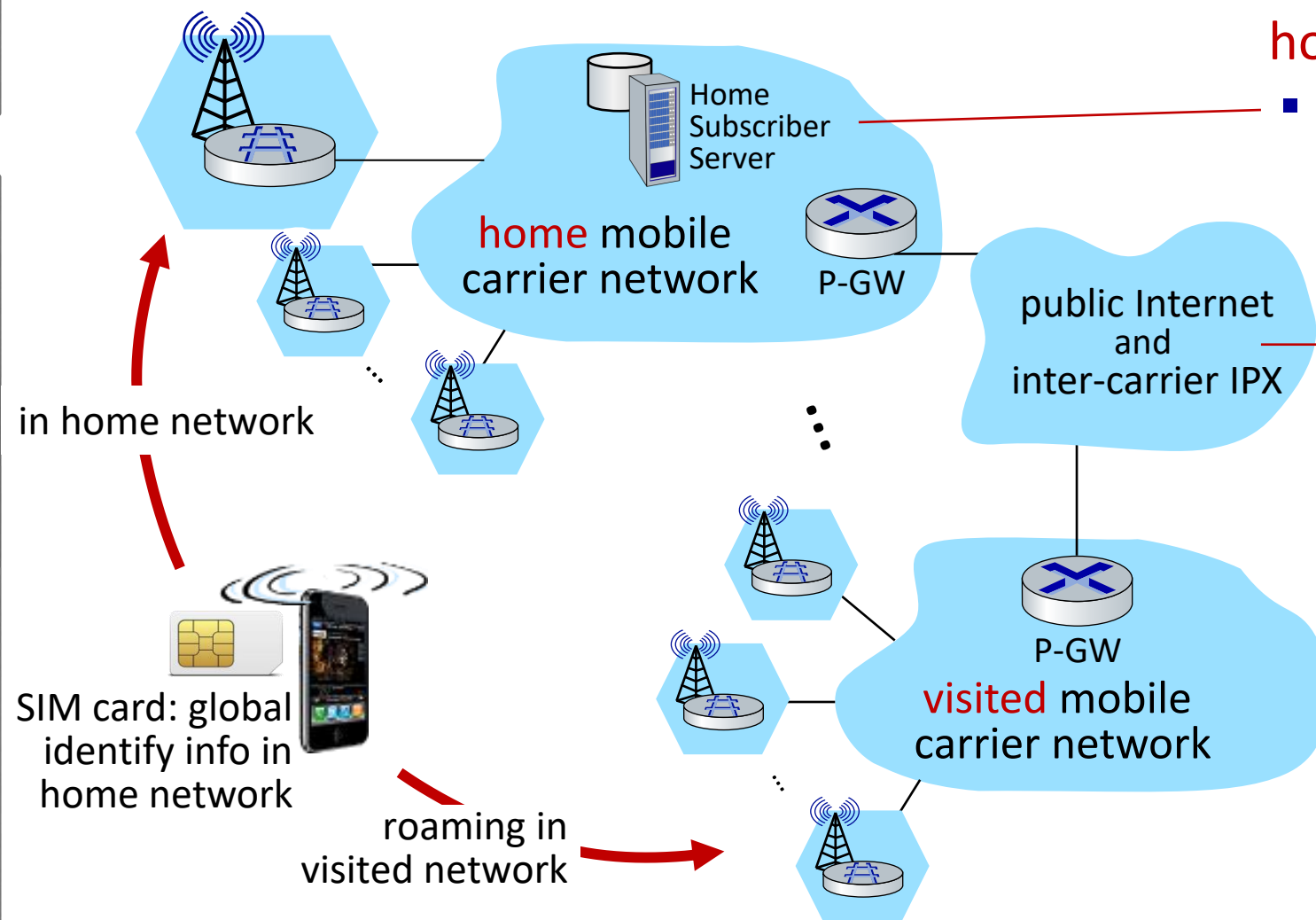
Global cellular network: a network of IP networks

7.1 无线与移动网络概述

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7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构



home network HSS:

- identify & services info, while in home network and roaming

all IP:

- carriers interconnect with each other, and public internet at exchange points
- legacy 2G, 3G: not all IP, handled otherwise



On to 5G!

7.1 无线与移动网络概述

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7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

- **goal:** 10x increase in peak bitrate, 10x decrease in latency, 100x increase in traffic capacity over 4G
- **5G NR (new radio):**
 - two frequency bands: FR1 (450 MHz–6 GHz) and FR2 (24 GHz–52 GHz): millimeter wave frequencies
 - not backwards-compatible with 4G
 - MIMO: multiple directional antennae
- **millimeter wave frequencies:** much higher data rates, but over shorter distances
 - pico-cells: cells diameters: 10-100 m
 - massive, dense deployment of new base stations required





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7.5 网络移动管理基本原理

聂兰顺



What is mobility?

❖ spectrum of mobility, from the **network** perspective:

no mobility

high mobility



device moves
between
networks, but
powers down
while moving

device moves
within same AP
in
one provider
network

device moves
among APs in
one provider
network

We're interested in these!

device moves
among multiple
provider
networks,
while maintaining
ongoing
connections





Contacting a mobile friend:

7.1 无线与移动网络概述

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7.4 移动通信网络体系结构

7.5 网络移动管理基本原理

Consider friend frequently changing locations, how do you find him/her?

- search all phone books?
- expect her to let you know where he/she is?

- call his/her parents?
- Facebook!

The importance of having a “home”:

- a definitive source of information about you
- a place where people can find out where you are

I wonder where Alice moved to?





Mobility: vocabulary

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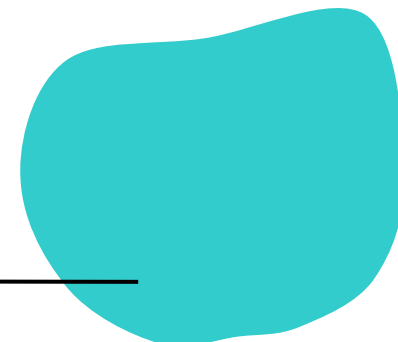
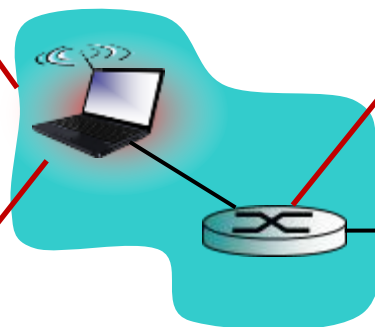
7.4 移动通信网络体系结构

7.5 网络移动管理基本原理

home network: permanent
“home” of mobile
(e.g., 128.119.40/24)

home agent: entity that will
perform mobility functions on
behalf of mobile, when mobile is
remote

permanent address:
address in home
network, can always be
used to reach mobile
e.g., 128.119.40.186



Mobility: vocabulary

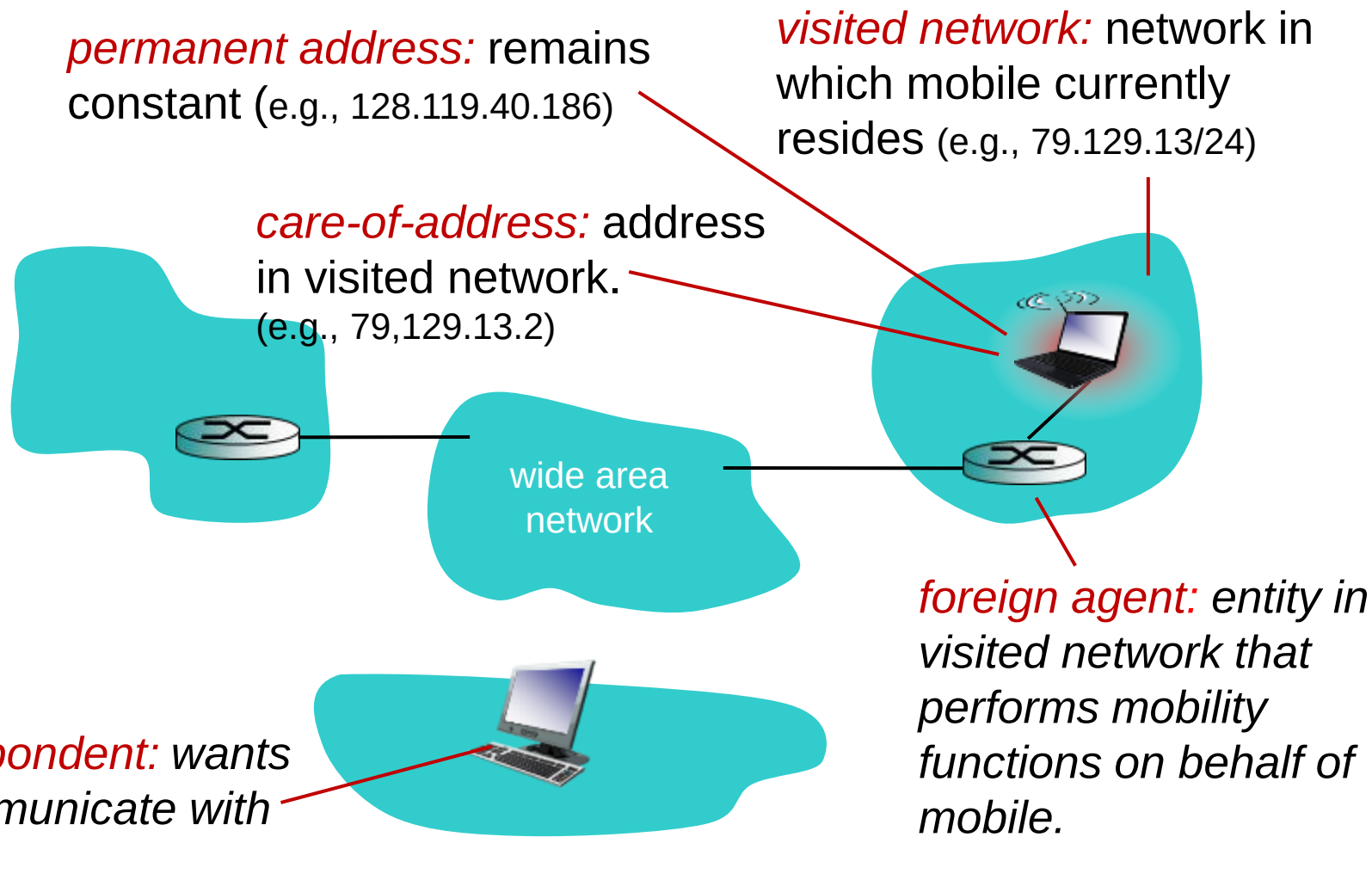
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Mobility approaches

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- 7.2 无线链路与网络特性
- 7.3 无线局域网:IEEE802.11
- 7.4 移动通信网络体系结构
- 7.5 网络移动管理基本原理**

❖ let network (routers) handle it:

- routers advertise well-known name, address (e.g., permanent 32-bit IP address), or number (e.g., cell #) of visiting mobile node via usual routing table exchange
- Internet routing could do this already *with no* changes! Routing tables indicate where each mobile located via longest prefix match!





Mobility approaches

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7.5 网络移动管理基本原理

■ let network (routers) handle it:

- routers advertise well known name, address (e.g., permanent 32-bit IP) or number (e.g., cell #) of visiting mobile in global routing table exchange
- Internet routing could already *with no changes!* Routing tables indicate where each mobile located via longest prefix match!

not
scalable
to billions of
mobiles

■ let end-systems handle it: functionality at the “edge”

- *indirect routing*: communication from correspondent to mobile goes through home network, then forwarded to remote mobile
- *direct routing*: correspondent gets foreign address of mobile, send directly to mobile





Mobility: registration

- 7.1 无线与移动网络概述
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7.5 网络移动管理基本原理



end result:

- foreign agent knows about mobile
- home agent knows location of mobile





Mobility via indirect routing

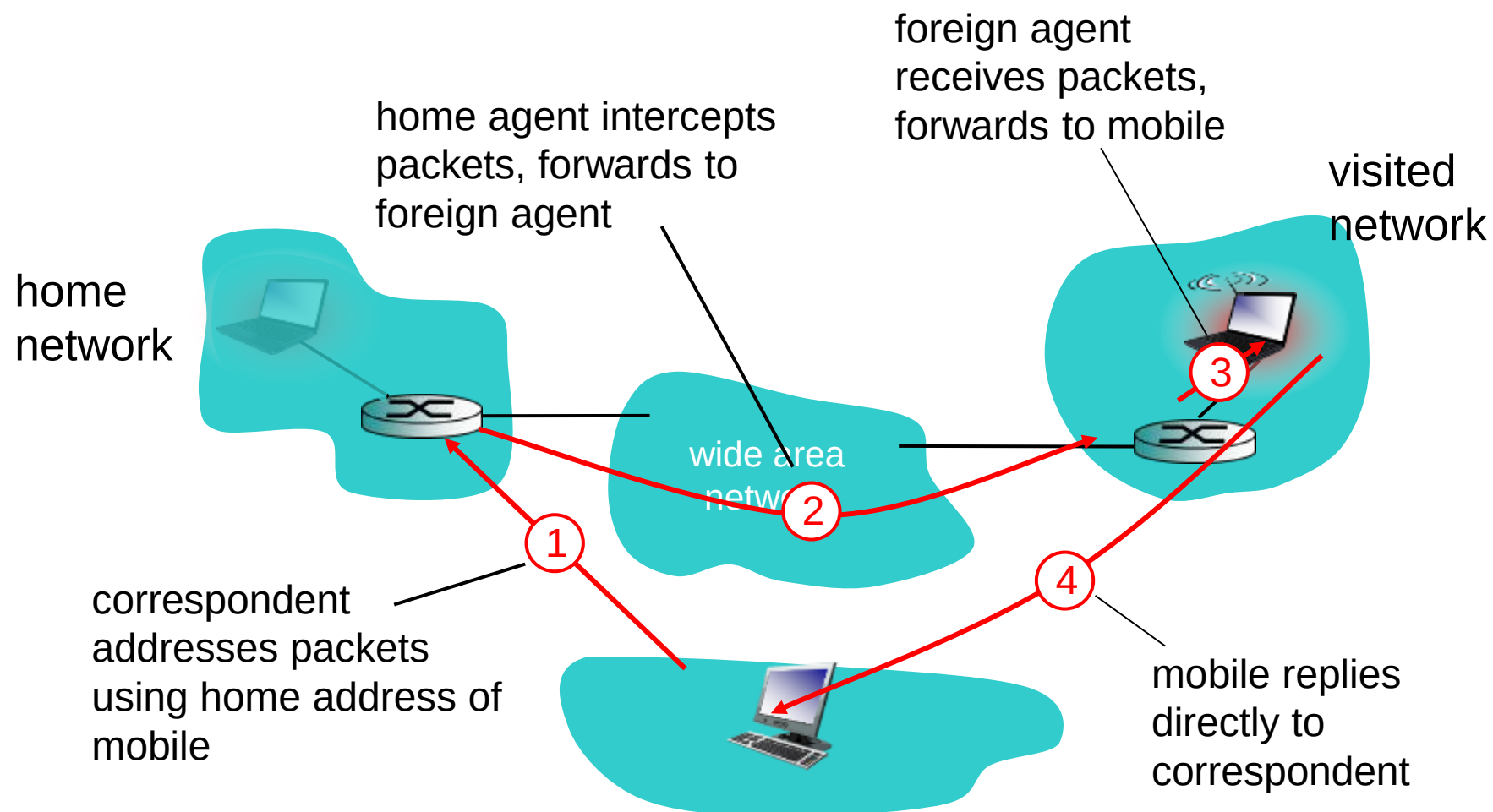
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Mobility via indirect routing

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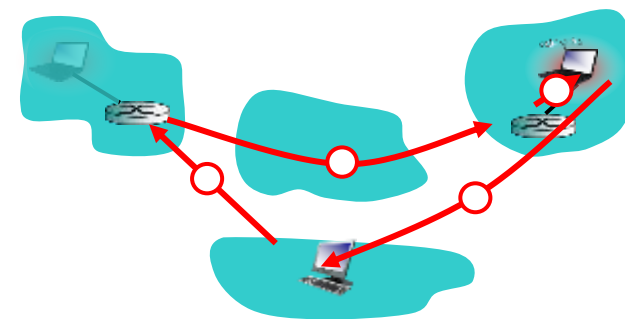
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7.4 移动通信网络体系结构

7.5 网络移动管理基本原理

- mobile uses two addresses:
 - **permanent address**: used by correspondent (hence mobile location is *transparent* to correspondent)
 - **care-of-address**: used by home agent to forward datagrams to mobile
- foreign agent functions may be done by mobile itself
- **triangle routing**: correspondent-home-network-mobile
 - inefficient when correspondent, mobile are in same network





Mobility via direct routing

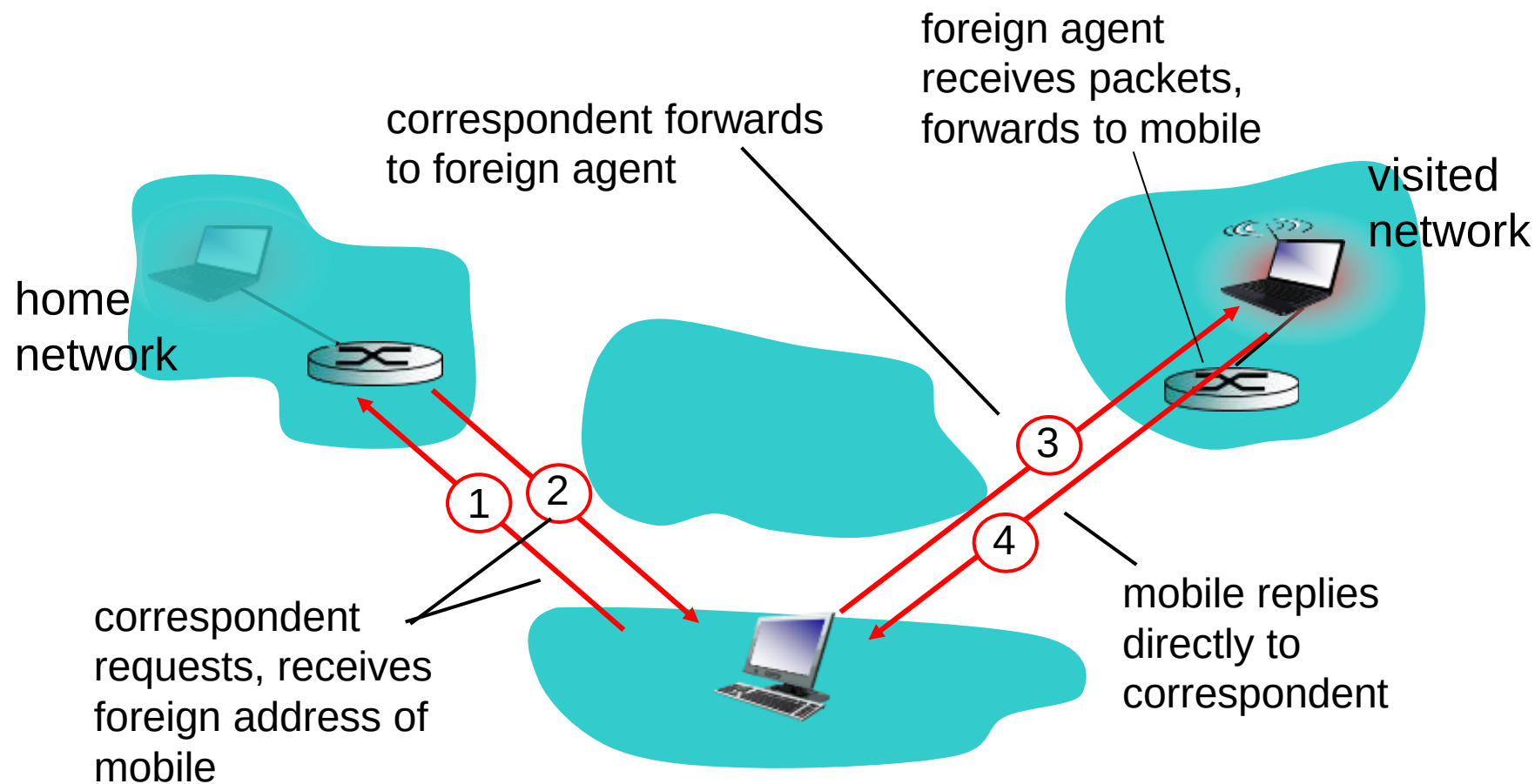
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7.2 无线链路与网络特性

7.3 无线局域网:IEEE802.11

7.4 移动通信网络体系结构

7.5 网络移动管理基本原理

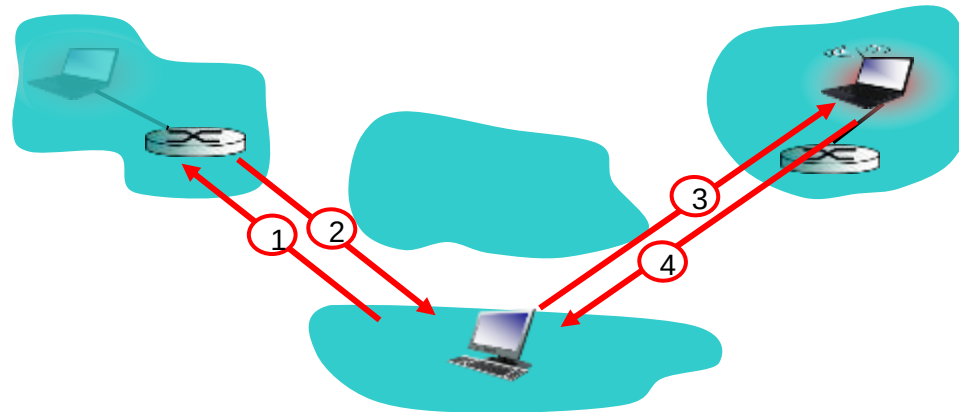




Mobility via direct routing

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- overcome triangle routing problem
- *non-transparent to correspondent:* correspondent must get care-of-address from home agent
 - what if mobile changes visited network?

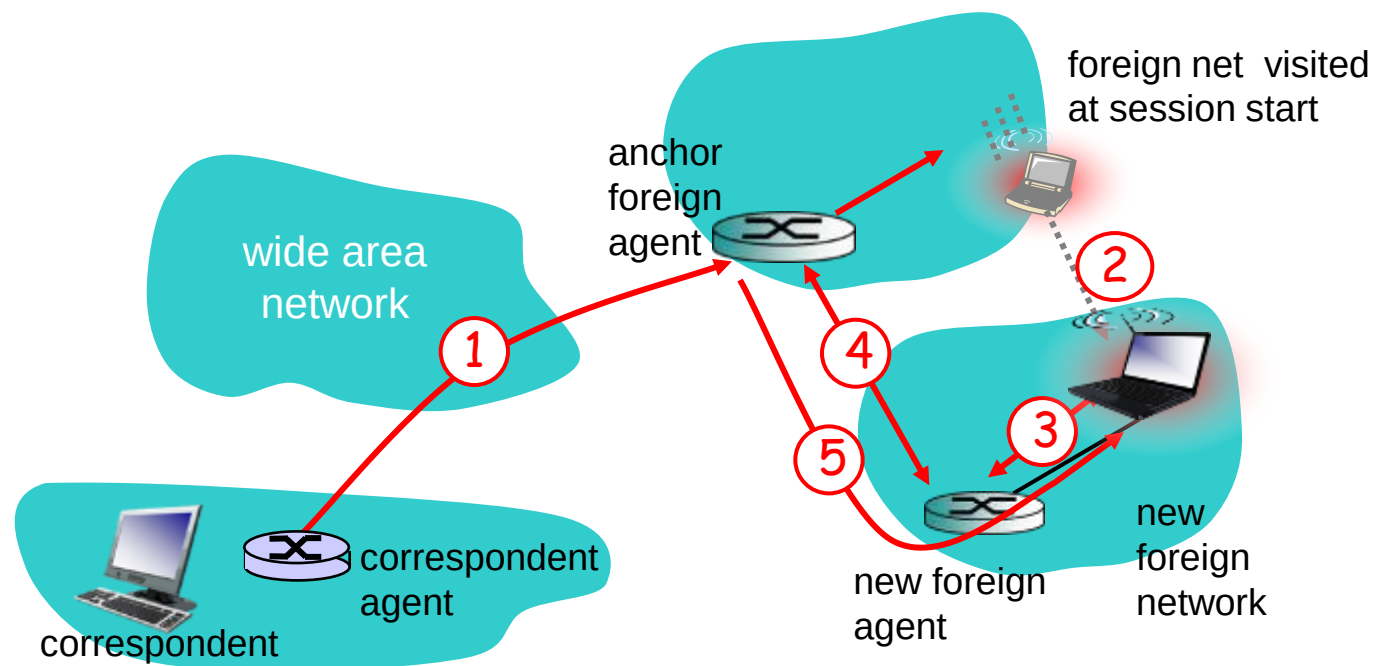




Accommodating mobility with direct routing

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- **anchor foreign agent**: FA in first visited network
- data always routed first to anchor FA
- when mobile moves: new FA arranges to have data forwarded from old FA (chaining)





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7.6 移动IP

聂兰顺



Mobile IP

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7.6 移动IP

❖ mobile IP architecture standardized ~20 years ago [RFC 5944]

- long before ubiquitous smartphones, 4G support for Internet protocols
- did not see wide deployment/use
- perhaps WiFi for Internet, and 2G/3G phones for voice were “good enough” at the time

❖ mobile IP architecture:

- indirect routing to node (via home network) using tunnels
- mobile IP home agent: combined roles of 4G HSS and home P-GW
- mobile IP foreign agent: combined roles of 4G MME and S-GW
- protocols for agent discovery in visited network, registration of visited location in home network via ICMP extensions





Mobile IP

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7.6 移动IP

❖ RFC 3344

❖ has many features we've seen:

- home agents, foreign agents, foreign-agent registration, care-of-addresses, encapsulation (packet-within-a-packet)

❖ three components to standard:

- indirect routing of datagrams
- agent discovery
- registration with home agent

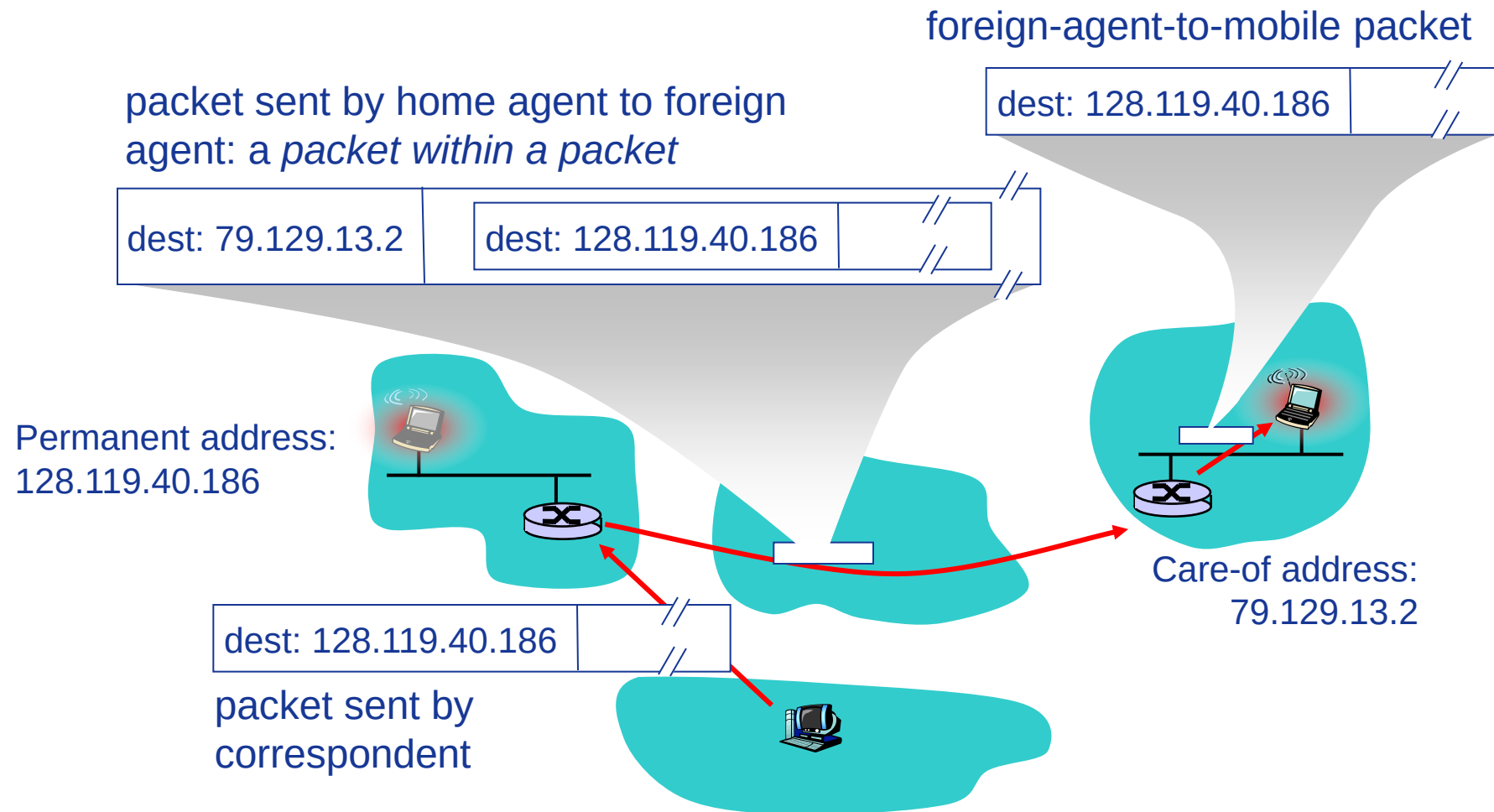




Mobile IP: indirect routing

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Mobile IP: agent discovery

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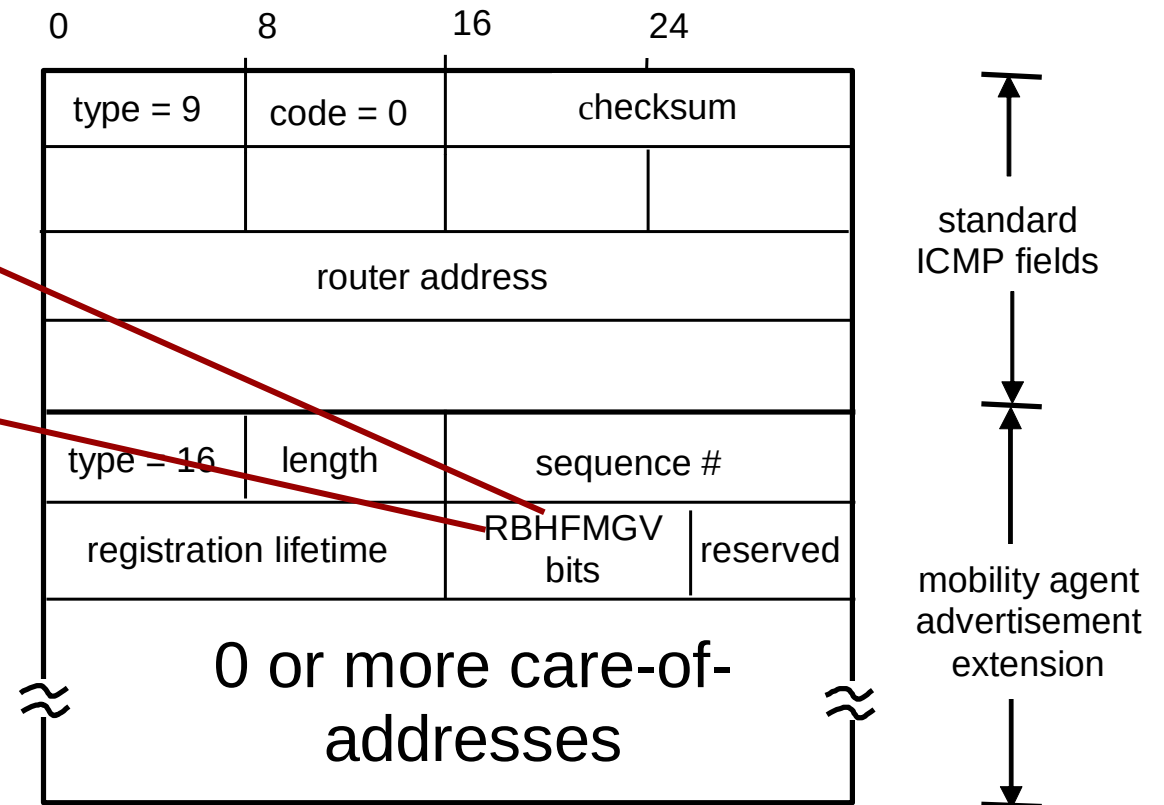
7.5 网络移动管理基本原理

7.6 移动IP

- ❖ *agent advertisement*: foreign/home agents advertise service by broadcasting ICMP messages (typefield = 9)

H,F bits: home and/or foreign agent

R bit: registration required





Mobile IP: registration example

7.1 无线与移动网络概述

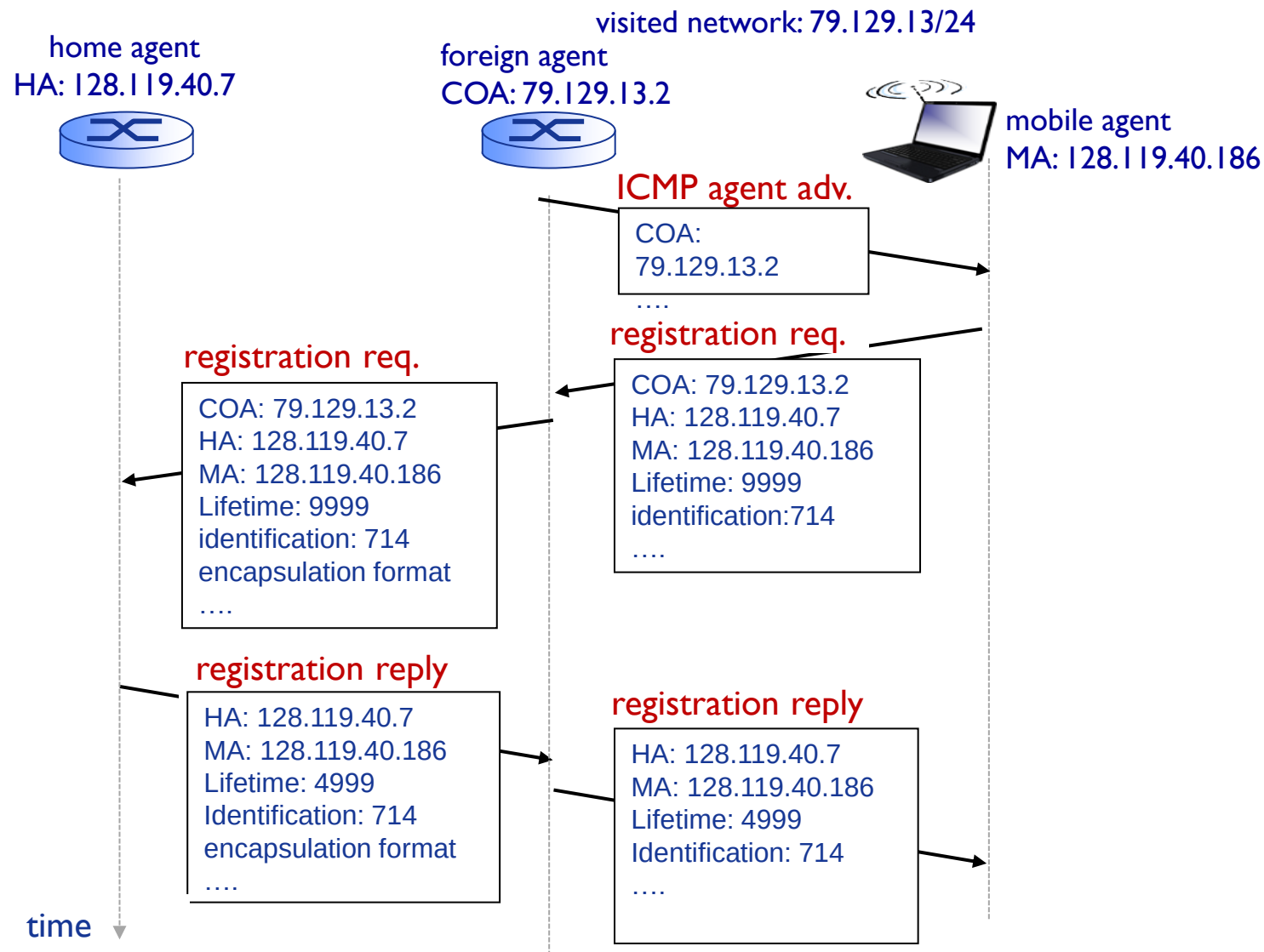
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7.7 移动通信网络的移动管理

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GSM: indirect routing to mobile

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7.7 移动通信网络的移动管理

❖ *home network*: network of cellular provider you subscribe to (e.g., Sprint PCS, Verizon)

- *home location register (HLR)*: database in home network containing permanent cell phone #, profile information (services, preferences, billing), information about current location (could be in another network)

❖ *visited network*: network in which mobile currently resides

- *visitor location register (VLR)*: database with entry for each user currently in network
- could be home network





GSM: indirect routing to mobile

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7.2 无线链路与网络特性

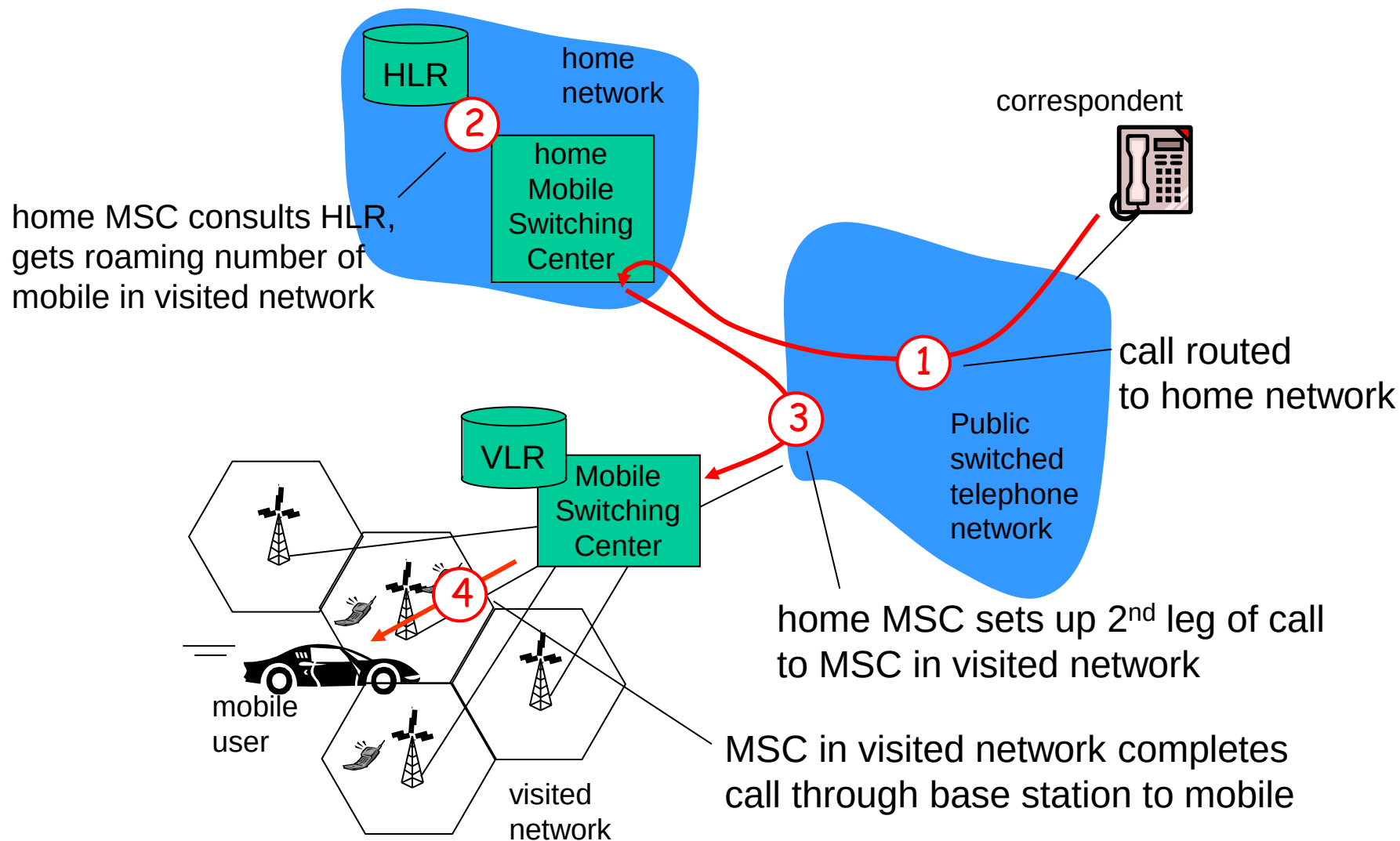
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7.7 移动通信网络的移动管理





GSM: handoff with common MSC

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7.2 无线链路与网络特性

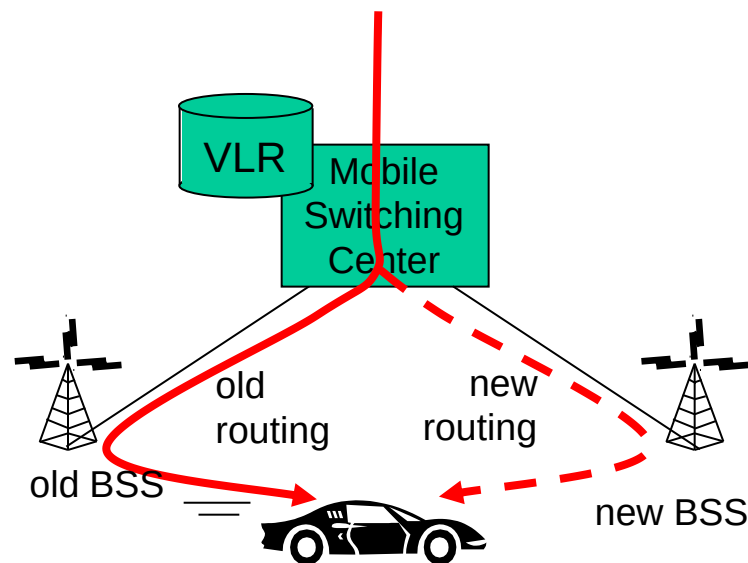
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7.7 移动通信网络的移动管理



- ❖ *handoff goal*: route call via new base station (without interruption)
- ❖ reasons for handoff:
 - stronger signal to/from new BSS (continuing connectivity, less battery drain)
 - load balance: free up channel in current BSS
 - GSM doesn't mandate why to perform handoff (policy), only how (mechanism)
- ❖ handoff initiated by old BSS

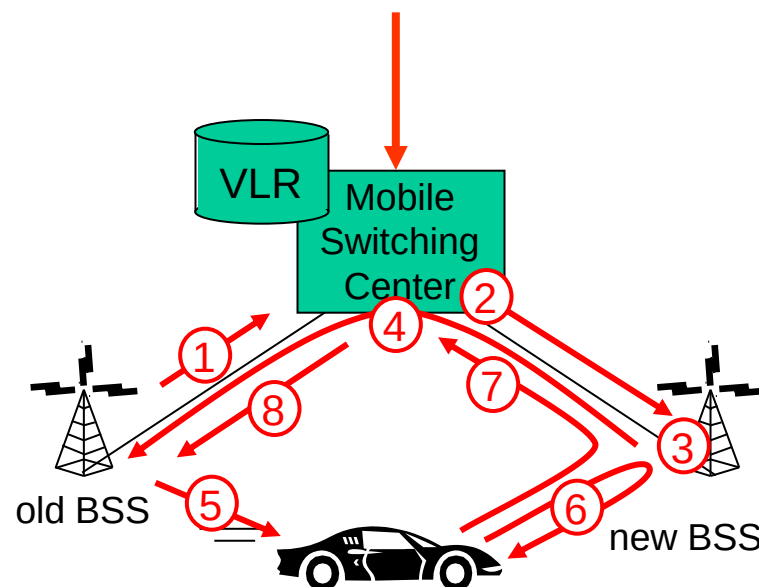




GSM: handoff with common MSC

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7.7 移动通信网络的移动管理



1. old BSS informs MSC of impending handoff, provides list of 1+ new BSSs
2. MSC sets up path (allocates resources) to new BSS
3. new BSS allocates radio channel for use by mobile
4. new BSS signals MSC, old BSS: ready
5. old BSS tells mobile: perform handoff to new BSS
6. mobile, new BSS signal to activate new channel
7. mobile signals via new BSS to MSC: handoff complete. MSC reroutes call
- 8 MSC-old-BSS resources released





GSM: handoff between MSCs

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7.2 无线链路与网络特性

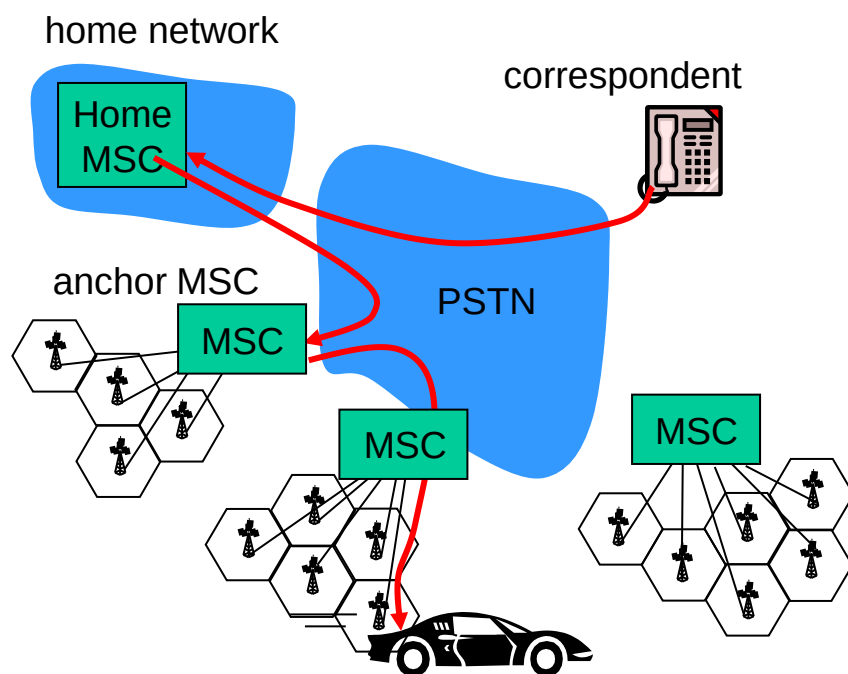
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7.7 移动通信网络的移动管理



(a) before handoff

- ❖ *anchor MSC*: first MSC visited during call
 - call remains routed through anchor MSC
- ❖ new MSCs add on to end of MSC chain as mobile moves to new MSC
- ❖ optional path minimization step to shorten multi-MSC chain

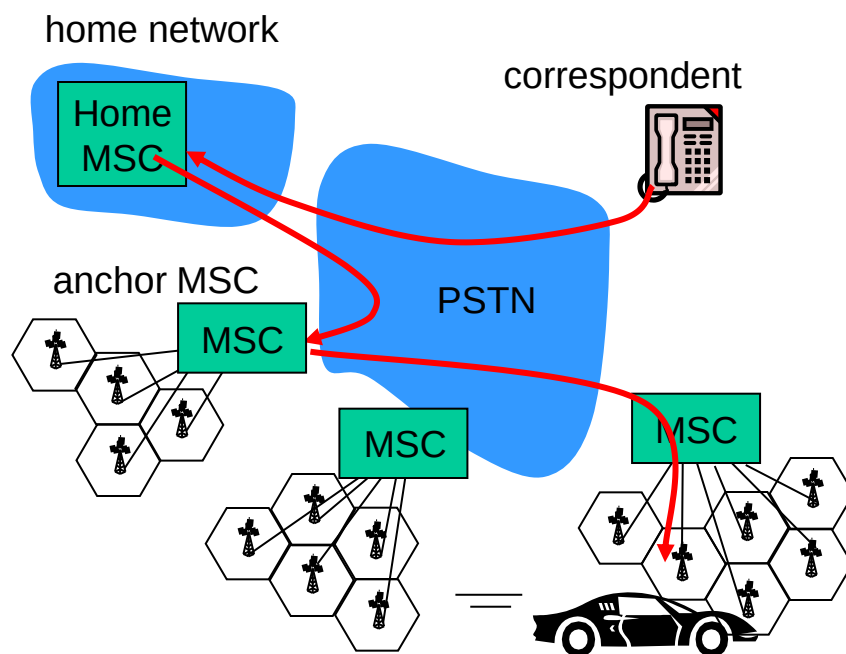




GSM: handoff between MSCs

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(b) after handoff

- ❖ *anchor MSC*: first MSC visited during call
 - call remains routed through anchor MSC
- ❖ new MSCs add on to end of MSC chain as mobile moves to new MSC
- ❖ optional path minimization step to shorten multi-MSC chain





Mobility: GSM versus Mobile IP

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7.7 移动通信网络的移动管理

GSM element	Comment on GSM element	Mobile IP element
Home system	Network to which mobile user's permanent phone number belongs	Home network
Gateway Mobile Switching Center, or "home MSC". Home Location Register (HLR)	Home MSC: point of contact to obtain routable address of mobile user. HLR: database in home system containing permanent phone number, profile information, current location of mobile user, subscription information	Home agent
Visited System	Network other than home system where mobile user is currently residing	Visited network
Visited Mobile services Switching Center. Visitor Location Record (VLR)	Visited MSC: responsible for setting up calls to/from mobile nodes in cells associated with MSC. VLR: temporary database entry in visited system, containing subscription information for each visiting mobile user	Foreign agent
Mobile Station Roaming Number (MSRN), or "roaming number"	Routable address for telephone call segment between home MSC and visited MSC, visible to neither the mobile nor the correspondent.	Care-of-address





Mobility in 4G networks: major mobility tasks

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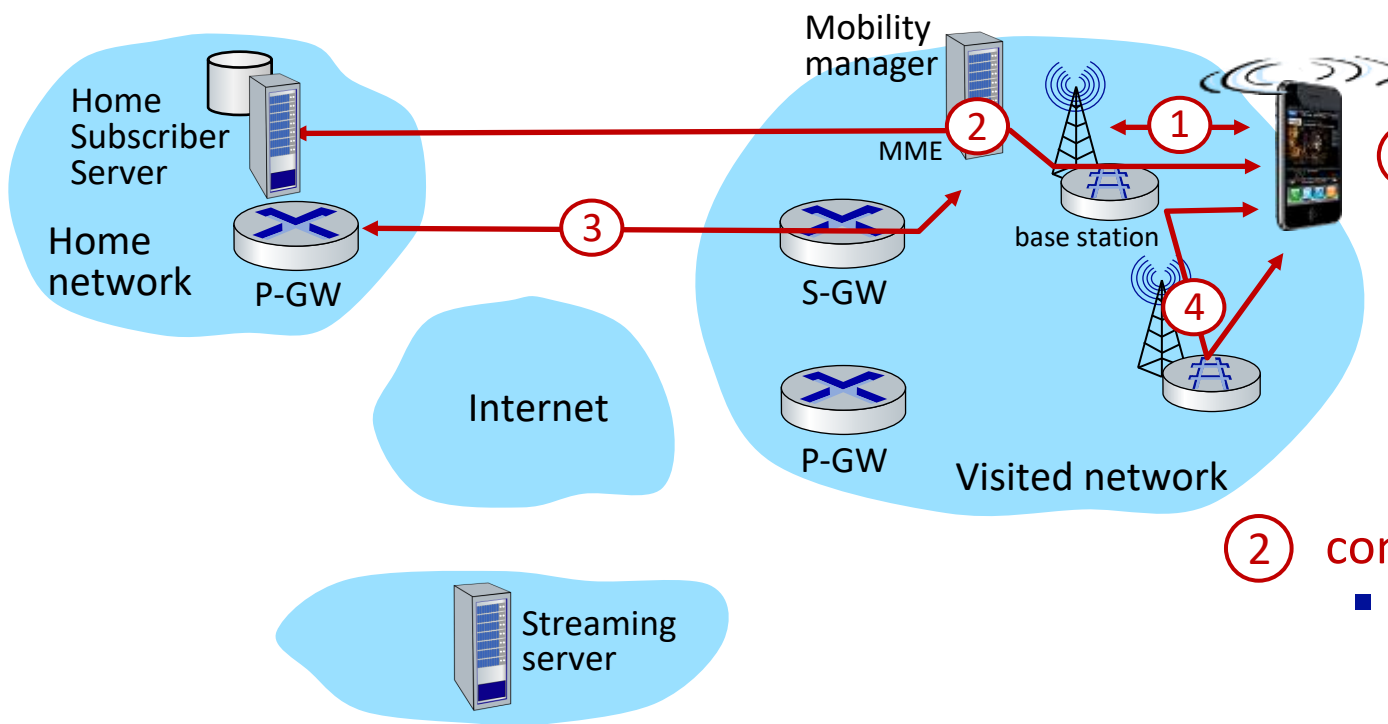
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7.7 移动通信网络的移动管理



① base station association:

- covered earlier
- mobile provides IMSI – identifying itself, home network

② control-plane configuration:

- MME, home HSS establish control-plane state - mobile is in visited network

③ data-plane configuration:

- MME configures forwarding tunnels for mobile
- visited, home network establish tunnels from home P-GW to mobile

④ mobile handover:

- mobile device changes its point of attachment to visited network





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7.8 无线与移动对高层协议的影响

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Wireless, mobility: impact on higher layer protocols

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7.8 无线与移动对高层协议的影响

❖ logically, impact *should* be minimal ...

- best effort service model remains unchanged
- TCP and UDP can (and do) run over wireless, mobile

❖ ... but performance-wise:

- packet loss/delay due to bit-errors (discarded packets, delays for link-layer retransmissions), and handover loss
- TCP interprets loss as congestion, will decrease congestion window un-necessarily
- delay impairments for real-time traffic
- bandwidth a scarce resource for wireless links





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